GAME DESIGN DOCUMENT

“OVERLINK”



FURRIPPL DIVISION PROJECT

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VERSION V0.17 (PRE-DEVELOPMENT DRAFT) - 16 OCT 2026

CHANGELOG

Date	Author	Changes	Version
23 Sep 2026	Muhamad Rizal	Organize initial GDD	v0.01
1 Oct 2026	Muhamad Rizal	Complete the system design, scopes and milestones	v0.02
15 Oct 2026	Muhamad Rizal	Require online sign-in for all progression; remove offline and anonymous accounts; define simultaneous victory resolution; split milestone verification lines	v0.03
15 Oct 2026	Muhamad Rizal	Restore anonymous guest sign-in; guest accounts may link an identity provider and keep their collection; offline accounts and offline progression remain unsupported	v0.04
15 Oct 2026	Muhamad Rizal	Replace Active/Bench/Ability card types with Unit/Ability (Active and Bench become board positions); define Bench promotion when the Active Unit is destroyed; make Bench passives and stat reading explicit	v0.05
15 Oct 2026	Muhamad Rizal	Fix residual combat contradiction: Bench Units are not destroyed with the Active Unit (align 2.1.6 with the 2.1.4 promotion rule); name the Active Unit in the base-damage formula	v0.06
16 Oct 2026	Muhamad Rizal	Development-readiness cleanup: normalize card terminology to Unit/Ability with Active/Bench as positions; fix lost apostrophes in 2.1.5; clarify Core Bench capacity in 2.1.3; state attack range explicitly in 2.1.6; fix validation wording in 4.15.4	v0.07
16 Oct 2026	Muhamad Rizal	Add Relay and Protocol card types (four types total): define placement, lifecycle, and deck-band share for all four types in 2.1.5 and 2.1.9; extend card anatomy, data-layer, and 4.15.4 validation rules to the four types	v0.08
16 Oct 2026	Muhamad Rizal	Clarify that Core and Node Bench slots may hold either a Bench Unit or Relay; define capacity, deployment, and occupied-slot	v0.09

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		rules accordingly; state that four deck type-share ranges are independent guidance and need not total 100%	
16 Oct 2026	Muhamad Rizal	Define [Lifesteal] as healing the damage-dealing card, not the player; replace HealCommander with Heal in the effect vocabulary	v0.10
16 Oct 2026	Muhamad Rizal	Resolve the effect-vocabulary mismatch: make the EffectType enum in 4.15.2 authoritative and align 4.4 with it (add DealAoEDamage, SwitchPosition, PreventBlock; rename ApplyBuff/ApplyDebuff to Buff/Debuff, DrawCards to Draw, MoveCard to Move, DestroyTarget to Destroy)	v0.11
16 Oct 2026	Muhamad Rizal	Add document versioning: a Version column in the CHANGELOG and the current version in the title block	v0.12
16 Oct 2026	Muhamad Rizal	Close review residuals on the effect vocabulary and versioning: 4.4 no longer conflates the 2.1.11 keyword glossary with the EffectType enum; add keyword granting to the vocabulary summary; state that v1.0 is assigned by a deliberate jump when the document is frozen	v0.13
16 Oct 2026	Muhamad Rizal	State that Commander and EnemyCommander address the player with no ATK, HP, or Shield; make Core durability the player's only health pool in 2.1.7; reject effects that require a card or health pool against Commander or EnemyCommander	v0.14
16 Oct 2026	Muhamad Rizal	Sync the title-block version with the CHANGELOG (it still read v0.12) and add an explicit 4.15.4 validation rule rejecting Commander or EnemyCommander targets that require a card or health pool	v0.15
16 Oct 2026	Muhamad Rizal	State that Self, Commander, and EnemyCommander all address the player (Self is an alias of Commander; EnemyCommander is the opponent's player), so no effect can be targeted at a Commander as if it were a card	v0.16
16 Oct 2026	Muhamad Rizal	Define the Core damage and repair rules:	v0.17

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		damage to a Core Active with no Shield points transfers 10% of the damage (rounded up) to its controller's Core as durability loss, in addition to the existing direct-attack and destruction events; add [Repair X] to restore Core durability, with Self and Commander as its legal targets	

Versioning: every applied revision increments the document version by one minor step and is listed in the Version column of the CHANGELOG above. Versions below v1.0 are pre-development drafts. Version v1.0 is the revision frozen for implementation: it is assigned by a deliberate jump when the document is frozen, not by the ordinary minor-step rule. Every revision after v1.0 increments by one minor step.

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1. INTRODUCTION

Overlink is a Trading Card Game (TCG) that combines strategic battlefield control, tactical card combat, and Active–Bench formations. Set in a fictionalized interpretation of Institut Teknologi Kalimantan (ITK), players deploy cards inspired by faculties, scientific fields, technology, and campus life across one central Core and four contested Nodes. Each battlefield features an Active Unit supported by Bench Units that provide buffs, passive effects, and strategic synergies. Players must balance combat, resource management, and territorial control to dominate the Nodes or exhaust their opponent's available Active Units. Overlink's unique gameplay focuses on the interaction between battlefield positioning, Active–Bench coordination, and intelligent decision-making through an AI-controlled opponent.

1.1. GAME CONCEPTS

The player takes on the role of a commander responsible for coordinating a faction's forces and making strategic decisions within a competitive network. The game focuses on the importance of coordination, adaptability, and resource management, where every decision can influence the balance of power across the battlefield. Through its ITK-inspired setting, Overlink explores the relationship between technology, innovation, and collaboration, emphasizing how different disciplines and capabilities can work together toward a shared objective.

1.2. GENRE

Digital Trading Card Game (TCG), Strategy, Tactical, Turn-Based

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1.3. TARGET AUDIENCE

Overlink is aimed at players aged 15 and above who enjoy strategy games, digital Trading Card Games, turn-based combat, and tactical decision-making. The game is primarily intended for casual and intermediate players who enjoy experimenting with card combinations, managing resources, and adapting their strategies during matches. It may also appeal to players interested in games with technology, academic, and campus-inspired themes.

1.4. PLATFORM

Overlink will be released for PC and Web, with distribution through itch.io. The game is designed to run on low-to-mid-range hardware to remain accessible to a broad range of players.

Specification	**Minimum**	**Recommended**
OS	Windows 10 or later	Windows 10 or later
Processor	Intel Core i3 / AMD equivalent	Intel Core i5 / AMD equivalent
RAM	4 GB	8 GB
Graphics	Integrated graphics supporting DirectX 11	NVIDIA GTX 1050 / AMD equivalent
Storage	2 GB available space	2 GB available space
Internet	PC single-player: Internet required for account sign-in, cloud saves, and all progression data	WebGL: Internet required to load the game, sign in, sync cloud saves, and access online services

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1.5. GAME CONTROLS

Overlink uses a mouse and keyboard control scheme. The mouse is the primary input for interacting with cards, selecting targets, navigating the battlefield, and activating actions. The keyboard is used for interface navigation and the listed menu and confirmation controls.

Input	**Action**
Left Mouse Button	Select cards, targets, and interface elements
Right Mouse Button	Cancel selection or return to the previous action
Esc	Open/close the pause or settings menu
Space	Confirm or proceed with the current action

1.6. FEATURES

Feature	**Description**
Strategic Digital TCG	Turn-based card battles focused on tactical decision-making and resource management
Core and Node Battlefield	A shared Core and four contested Nodes create multiple objectives within a single match
Active-Bench System	Active Units engage directly in combat

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	while Bench Units provide support, passive effects, and synergies
Faction-Based Gameplay	Different factions offer distinct playstyles, card abilities, and strategic approaches
ITK-Inspired Card Collection	Cards are inspired by faculties, technology, academic fields, campus environments, and student life at Institut Teknologi Kalimantan
Card Collection and Gacha	Players can collect cards through randomized card packs featuring different rarities and card types
Node Domination	Players can pursue victory by strategically controlling the battlefield's Nodes
Card Synergy	Combining specific Active and Bench Units can unlock additional effects and tactical advantages
AI Opponent	Single-player matches feature an AI-controlled opponent that analyzes the current game state and makes strategic decisions during gameplay
Online Multiplayer	Deferred beyond MVP: players may compete in online matches after a later scope and production review
PC and Web Support	Playable on PC and directly through a web browser via itch.io

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1.7. LOOK AND FEEL

A near-future network contest pairs painterly character art with a geometric, scan-line world. Dark grounds and restrained faction-colour signals make battlefield control legible at a glance; see 7.1–7.2 for the art direction and visual rules.

1.8. COMPETITIVE ANALYSIS

Game	Similarity	Differences
Legends of Runeterra	Turn-based card combat, card abilities, resource management, and strategic card combinations.	Overlink introduces a multi-battlefield structure with a shared Core and four Nodes, as well as an Active-Bench system.
Pokémon Trading Card Game	Active and Bench Units, card collection, card abilities, and switching between Active and Bench positions.	Overlink focuses on controlling multiple Nodes and competing for a shared Core, while its Bench system emphasizes battlefield-specific support and synergies.

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2. GAMEPLAY MECHANICS

2.1. CORE MECHANICS

2.1.1. RESOURCE SYSTEM: NETWORK POINTS (NP)

Network Points (NP) represent the player's available bandwidth and processing power to deploy and maintain cards on the Overlink network.

Core Rules:

- Each player starts with 3 NP at the beginning of the game
- At the start of each turn, gain +1 NP (up to a maximum of 10 NP)
- Node Control Bonus: Each Node you control grants +1 NP at turn start (stacks with base gain) Example: Control 2 Nodes → gain 3 NP per turn (1 base + 2 from Nodes)
- NP is persistent — unspent NP carries over to the next turn
- Deploying cards, activating abilities, and switching Active/Bench positions cost NP

Design Rationale:

- Incentivizes Node control (territorial objective feeds economic advantage)
- Creates catch-up mechanic (losing player gets base NP, winning player must maintain Nodes)
- Persistent NP rewards conservative play but risks falling behind on tempo

2.1.2. TURN STRUCTURE

Each turn follows 6 phases executed in order:

Phase 1: Upkeep

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- Gain NP: +1 base + +1 per controlled Node
- Check NP cap (max 10 NP)
- Resolve "at start of turn" card effects
- Check victory conditions (if any are met, game ends immediately)

Phase 2: Draw

- Draw 1 card from your deck
- Deck-Out: Opponent cannot draw a card during their Draw phase and has no cards in hand

Phase 3: Main Phase

- Deploy cards to available Active or Bench positions
- Activate [Activate] abilities during your Main Phase. [Fast] cards/abilities may be used only by the non-active player between resolved commands during the active player's Main Phase (2.1.11). Resolve immediately; no nested responses or other-phase windows.
- Switch Active ↔ Bench positions (costs NP, limit 1 per battlefield per turn)
- Order of actions is player's choice

Phase 4: Combat

- Declare Attacks: Choose which Active Units attack enemy Active Units or the enemy Core. Nodes themselves cannot be attacked; control follows occupation (2.1.7).
- Declare Blocks: Opponent assigns blockers from their Active Units
- Resolve Combat: Calculate damage, apply effects, destroy cards with 0 or less HP
- Combat is simultaneous (attacker and blocker both deal damage)

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Phase 5: End Phase

- Resolve "at end of turn" card effects
- Check for destroyed cards (move to discard pile)
- Check Node/Core control status
- Buff/debuff durations decrement by 1

Phase 6: Cleanup

- Empty temporary zones (e.g., "this turn" effects expire)
- Pass turn to opponent

2.1.3. BATTLEFIELD STRUCTURE

Zones:

- Core (shared):
  - Each player has 3 Core Active slots and 3 Core Bench slots, shared among that player's Core Actives. Each Core Bench slot may hold one Bench Unit or one Relay. Each Bench Unit is assigned to support one friendly Core Active; a Relay occupies a slot but is not assigned and grants no Bench Unit passive. Bench capacity is not multiplied by the number of Active slots.
  - Core provides multiplier effects based on controlled Nodes (see 2.1.7)
- Nodes (contested):
  - Each Node has 1 Active slot + 2 Bench slots per player; each Bench slot may hold one Bench Unit or one Relay
  - Nodes are neutral at game start; control is contested through card presence

Deck & Hand:

- Deck: 30–40 cards (player chooses size within range)
- Starting Hand: Draw 5 cards at game start

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- No mulligan in MVP (defer to post-MVP)
- Hand Size: No maximum. Hand size is limited only by deck-out.

2.1.4. ACTIVE & BENCH SYSTEM

Active Units:

- Positioned in Active slots (Core or Node)
- Can attack enemy Active Units or the enemy Core; Nodes are controlled by occupation, not damage.
- Can block enemy attacks
- Visible to opponent (stats, abilities, buffs shown)
- Take damage from combat and abilities

Bench Units:

- Positioned in Bench slots (behind an Active Unit in same battlefield)
- Cannot attack or block
- Cannot be directly targeted by enemy cards or abilities (except AoE or "target any card" effects)
- Provide passive buffs to the Active Unit in their battlefield (a Bench Unit's passive is active while it is Benched):
    - Stats: +ATK, +HP, +Shield
    - Keywords: Grant abilities like [Piercing], [Heal], [Draw]
    - Triggered effects: "When your Active attacks...", "At start of turn..."
- Promotion:
    - If the Active Unit in a battlefield is destroyed, the Bench Units in that battlefield are not destroyed. The controlling player chooses one Bench Unit in that battlefield and promotes it into the now-empty Active slot. Promotion costs no NP and does not trigger Summoning Delay. If no Bench Unit remains in that battlefield, the Active slot stays empty

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- If Active switches out, Bench Units stay in that battlefield (remain Benched for the new Active)

Switching (Active ↔ Bench):

1. Cost: 2 NP per switch
2. Limit: Once per battlefield per turn (can switch Core, N1, N2, N3, N4 separately)
3. Restriction: Cannot switch a card that was deployed this turn (Summoning Delay — see 2.1.5)
4. Process:
   - 4.1. Pay 2 NP
   - 4.2. Select Active Unit to move to Bench
   - 4.3. Select Bench Unit to move to Active
   - 4.4. Swap positions immediately (no stack/response window in MVP)

Bench Capacity:

- Core: Max 3 Bench slots total per player, shared among that player's Core Actives; each slot holds one Bench Unit or Relay.
- Node: Max 2 Bench slots total per player at each Node; each holds one Bench Unit or Relay, and each Bench Unit supports that player's sole Active slot there.
- If all Bench slots are occupied, cannot deploy an additional Bench Unit or Relay to that battlefield unless a slot opens

2.1.5. DEPLOYMENT RULES

Deploying Cards by Type:

Card Types (MVP):

Unit: a combat card deployed to an Active or Bench position. Active Units may attack and block; Bench Units cannot attack or block and provide their passive support to the friendly Active they support at that battlefield. A Bench Unit may be promoted under 2.1.4.

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Ability: a one-shot card effect. It occupies no board slot, resolves immediately when legally played, then enters the discard pile unless [Exhaust] removes it from the game. [Fast] timing follows 2.1.11.
Relay: a face-up, persistent noncombat card deployed to an empty Bench slot at the Core or a Node. It consumes one shared Bench slot, cannot attack, block, or count for Node control, and has no ATK, HP, or Shield. Its printed effect applies to its controller's Active Unit at that same battlefield while the Relay remains there; if no friendly Active is present, the effect is dormant. In the MVP, combat does not damage a Relay and it cannot be destroyed by damage; it remains in its Bench slot until the match ends. Example: "Aegis Relay" (cost 2) — "Your Active Unit at this battlefield gets +1 Shield."
Protocol: a face-up, persistent player-side card in a dedicated Protocol slot. It consumes no battlefield slot and cannot attack, block, or count for Node control. Each player may control at most one Protocol. Playing a new Protocol replaces the old one: after the new card's cost is paid and it is placed, the previous Protocol is moved to its owner's discard pile. A Protocol effect applies only to its controller and stops when the card leaves play. Protocols may be played only during their controller's Main Phase. Example: "Open Channel" (cost 3) — "At the start of your Upkeep, gain 1 NP, up to the 10 NP cap (2.1.1)."
Abilities resolve and leave play; Units and Relays occupy battlefield positions; Protocols use the Protocol slot. All four types are face-up. Only Units use combat stats; the stat fields of Ability, Relay, and Protocol cards are zero. Examples are illustrative rules text, not separate serialized effect names.

1. Pay the card's NP cost (shown in the top-left corner)
2. Unit: choose an available Active or Bench slot at the Core or a Node. Relay: choose an empty Bench slot at the Core or a Node. Ability: uses no board slot; play during your Main Phase or a legal [Fast] window (2.1.11), resolve immediately, then discard unless [Exhaust] applies. Protocol: play into your single Protocol slot during your Main Phase only.
3. All card types are played face-up and visible to both players

Deployment Restrictions:

- Summoning Delay: Cards deployed this turn cannot attack or switch until your next turn
- Battlefield Availability:

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    - Core: Always available for deployment
    - Nodes: Deploy an Active into any empty friendly Active slot, including at a neutral or enemy-controlled Node. Deploy a Bench Unit or Relay only if you have an Active in that Node and an empty friendly Bench slot. Pay normal costs and apply Summoning Delay to deployed Units; Relays cannot attack or switch. A card deployed by [Fast] enters its controller's battlefield but cannot attack or switch until the start of that controller's next turn.
- Faction Limits: (MVP: single-faction decks only — mixed factions deferred to post-MVP)

Special Deployment:

- Some cards have [Deploy: Effect] abilities that trigger when deployed
- Some cards or abilities carry the [Fast] keyword, allowing the non-active player to use them between resolved commands during the opponent's Main Phase (2.1.11).

2.1.6. COMBAT RESOLUTION

Combat Flow:

Step 1: Declare Attacks

- Active player chooses which Active Units attack. An Active Unit may attack any enemy Active Unit or the enemy Core, regardless of which battlefield either occupies
- For each attacker, declare target:
    - Enemy Active Unit (direct combat)
    - Nodes are not attack targets. Attacking an enemy Active at a Node does not move the attacker there or grant control without friendly occupation (2.1.7).
    - Enemy Core (direct attack; the defending player may assign eligible Core Actives to block in Step 2)

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Step 2: Declare Blocks

- Defending player may assign eligible Active Units in the same battlefield as the target, at most one blocker per attacker and one attacker per blocker. Blocking is optional; [Cannot Block] cards cannot block. Only Core Actives may block a direct Core attack.
- One blocker per attacker (no multi-block in MVP)
- A blocked attacker fights its blocker and deals no direct damage to its declared target. An unblocked attacker hits its declared target.

Step 3: Resolve Damage

- Combat is simultaneous (both attacker and blocker deal damage)
- Damage Calculation:
  - Base damage = the Active Unit's ATK stat
  - Modified by buffs/debuffs from Bench Units, abilities, and effects
- Damage Assignment:
  - Attacker deals ATK damage to blocker (reduces blocker's HP)
  - Blocker deals ATK damage to attacker (reduces attacker's HP)
- Destruction Check:
  - If HP ≤ 0, card is destroyed (moved to discard pile at end of combat)
  - If the destroyed card was the Active Unit of its battlefield, that battlefield's Bench Units are not destroyed; promote one into the empty Active slot (2.1.4)

Step 4: Unblocked Damage

- Each unblocked direct attack on the enemy Core reduces Core durability by exactly 1. Separately, each Core Active destroyed

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by combat or an effect reduces its owner's Core durability by exactly 1, even if it was a blocker. These are distinct events and may both occur in one combat. Nodes have no damage or control counter; resolve attacks against their Active Units using normal combat. Damage dealt to a Core Active Unit with no Shield points also transfers to its controller's Core (2.1.7).

Combat Keywords:

- [Piercing]: Damage ignores shields
- [First Strike]: Deals damage before opponent (opponent only retaliates if survives)
- [Lifesteal]: Heal the card that dealt the damage for that amount.
- [AoE X]: Deals X damage to all enemy Active Units in the same battlefield
- [Cannot Block]: This card cannot be assigned as a blocker

2.1.7. NODE & CORE CONTROL

Node Control Mechanics:

Node States:

- Neutral: Neither player has an Active in the Node (all Nodes start empty and neutral). Bench Units alone do not establish control.
- Contested: Both players have an Active in the Node. Neither player controls it or receives its control benefits.
- Controlled: Exactly one player has an Active in the Node; that player controls it.

Gaining Control:

1. Deploy an Active into the Node's empty friendly Active slot under 2.1.5. This may establish control or contest the opponent's control.

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2. After each resolved action or simultaneous damage/destruction batch, recompute control from surviving Active presence. If only your Active is present, you immediately control the Node, even during Summoning Delay.
3. If both players have an Active, the Node is contested. Destroying an enemy Active from another battlefield does not move your attacker into the Node or grant control there.

Losing Control:

- Control ends immediately when your last Active leaves the Node or an opposing Active enters it; do not wait until End Phase.
    - There is no attrition counter, Node damage threshold, or retained ownership without an Active. Recompute after the complete action or simultaneous batch, not between the two positions of an atomic switch.
- If only the opponent has an Active, they control the Node; if neither player has one, it is neutral. Apply or remove control benefits immediately. Already gained NP is not reversed.

Node Benefits:

- Each controlled Node grants +1 NP per turn
- Each controlled Node provides a Core Multiplier (see Core mechanics below)

Core Control Mechanics:

Core Damage:

- Your Core has a durability counter starting at 3. Core durability is the only health pool a player owns: the Commander is the player as addressed by effects, not a card with ATK, HP, or Shield (4.15.2)

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- Each time one of your Core Active Units is destroyed by combat or an effect, your Core durability decreases by 1. This is separate from direct Core damage: an unblocked attacker reduces durability by 1, while a blocker's destruction separately reduces it by 1. Count each event once; a blocked attack causes no direct Core damage.
- Damage transfer from Core Actives: when a Core Active Unit with no Shield points takes damage, 10% of that damage is transferred to its controller's Core as durability loss, rounded up per instance (1 to 10 damage costs 1 durability, 11 to 20 costs 2, and so on). A Core Active has no Shield points when its Shield stat is 0 or its Shield was already consumed. Only Core Active Units transfer damage this way; Actives at a Node do not. This is separate from the destruction and direct-attack losses above and is counted once per instance.
- When durability reaches 0, you lose the game

Core Multipliers (Node Synergies):

- Each controlled Node grants a buff to all Active Units in your Core:
    - Node 1: +1 ATK to Core Active Units
    - Node 2: +1 HP to Core Active Units
    - Node 3: Core Active Units gain [First Strike]
    - Node 4: Core Active Units gain [Draw 1 card when they destroy an enemy]
- Multipliers stack (control multiple Nodes → stack multiple buffs)
- Buffs are active only while Node is controlled (lost immediately if Node is lost)

Core Defense Priority:

- Core attacks may be declared whether or not blockers are present. During Step 2 of 2.1.6, the defender may assign Core Actives to block; each unblocked direct Core attack causes

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exactly 1 durability loss. A Core Active destroyed in that combat causes its separate 1-point loss as described above.

2.1.8. VICTORY CONDITIONS

A player wins immediately when any of the following occur:

1. Node Domination: Control 3 out of 4 Nodes at the start of your Upkeep phase
2. Core Destruction: Opponent's Core durability reaches 0
3. Deck-Out: Opponent cannot draw a card during their Draw phase and has no cards in hand
4. Concession: Opponent concedes the match

Simultaneous victory resolution: resolve every effect in the current atomic batch before checking victory conditions. If both players satisfy a victory condition in that same batch, the active player (whose turn it is) wins. If only one player satisfies a condition in the batch, that player wins.

-

2.1.9. DECK CONSTRUCTION RULES

Deck Requirements:

- Deck Size: 30–40 cards (player chooses)
- Card Copies: Max 3 copies of any single card
- Faction Restriction (MVP): All cards must belong to the same faction

Post-MVP: Allow 2-faction "splash" decks with restrictions

Deck Composition Guidelines (not enforced, but recommended):

The four type bands below are independent recommendations, not validation rules, so a deck need not satisfy all four or form a complete 100% partition. Percentages are of final deck size; the reference deck below is one exact composition.

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- Unit cards (Active combat and Bench support roles): 50–60%
- Ability cards (one-shot effects): 15–25%
- Relay cards (persistent battlefield support; occupy a Bench slot): 10–15%
- Protocol cards (persistent player-side effects; occupy the Protocol slot): 10–15%

Reference deck, 30 cards: 16 Units (53%), 6 Abilities (20%), 4 Relays (13%), and 4 Protocols (13%). These are a balancing reference, not a legality requirement; new faction lists are balanced against this distribution.

Rarity Distribution (for collection balance):

The MVP library contains exactly 100 card definitions: 50 Common, 30 Rare, 15 Epic, and 5 Legendary. These are collection counts, not pack probabilities. Each earned-currency MVP pack contains 5 cards; each slot independently rolls rarity at Common 50%, Rare 30%, Epic 15%, or Legendary 5%, then selects uniformly among valid definitions in that tier. Duplicate results are allowed and convert to shards under 4.12. Pack opening requires online service access; offline play cannot open packs. MVP includes no real-money purchase or paid currency.

Rarity	**Count**	**Share**
Common	50	50%
Rare	30	30%
Epic	15	15%
Legendary	5	5%

The exact library counts require no rounding. The pre-production vertical slice ships 80 card definitions at the same ratio: 40/24/12/4. Pack odds remain 50/30/15/5 for that slice; random packs do not guarantee a complete collection.

2.1.10. CARD ANATOMY TEMPLATE

Every card is defined by the same set of fields.

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Field	Description
Cost	NP required to deploy
Type	Unit, Ability, Relay, or Protocol. A Unit occupies an Active or Bench position; an Ability is a one-time effect that occupies no slot; a Relay is a persistent noncombat card in a Bench slot; a Protocol is a persistent player-side card in the Protocol slot. See 2.1.5.
Stats	ATK (damage dealt), HP (health), Shield (absorbs damage before HP is reduced)
Keywords	Standardized mechanics (see 2.1.11)
Ability Text	Unique effects using standardized templates

2.1.11. KEYWORD GLOSSARY (MVP SET)

Keyword	Effect
[Piercing]	Damage ignores Shield values
[First Strike]	Deals combat damage before opponent. If opponent survives, they retaliate
[Lifesteal]	When this card deals damage, heal this card for that amount.
[Draw X]	Draw X cards from your deck
[AoE X]	Deal X damage to all enemy Active Units in the same battlefield
[Cannot Block]	This card cannot be assigned as a

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	blocker during combat
[Activate: Cost]	Pay the cost during your Main Phase to trigger this effect
[Deploy: Effect]	Trigger this effect immediately when this card is deployed
[Fast]	Only the non-active player may use this card/ability in the opponent's Main Phase, between resolved commands. At most one [Fast] use is allowed per resolved command; it resolves immediately and creates no nested response. No [Fast] use is allowed during other phases or combat resolution.
[Shield X]	This card has X Shield points that absorb damage before HP is reduced
[Heal X]	Restore X HP to target card (cannot exceed max HP)
[Summoning Delay]	Cannot attack or switch the turn it is deployed
[Persistent]	This buff/debuff does not expire at end of turn (must be removed by effect)
[Exhaust]	After using this card/ability, remove it from the game (not discarded)
[Repair X]	Restore X Core durability to your Core (cannot exceed the starting value of 3)

2.1.12. EXAMPLE TURN WALKTHROUGH

Setup

- Player A (you) controls Node 1 and Node 2

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- Player B (opponent) controls Node 3
- Both players have 3 Active Units in their Core
- This is a mid-game turn rather than the opening turn: both players begin the turn holding 5 NP and a hand of 5 cards

Your Turn Begins

Phase 1: Upkeep

- Gain NP: 1 (base) + 2 (Nodes 1 & 2) = 3 NP → Total NP: 8
- Check buffs: Your Core Active Units have +1 ATK and +1 HP (from Nodes 1 & 2)

Phase 2: Draw

- Draw 1 card → Hand size: 6 cards

Phase 3: Main Phase

- Action 1: Deploy a Bench Unit (cost 2 NP) to support your Core Active → NP: 6
- Action 2: Activate an ability on a Bench Unit (cost 3 NP) to buff your Active in Node 1 → NP: 3
- Action 3: Switch your Active in Node 2 with a Bench Unit (cost 2 NP) → NP: 1

Phase 4: Combat

- **Declare Attacks:**
  - Core Active #1 attacks enemy Core Active #1
  - Node 1 Active attacks enemy Node 3 Active
  - Node 2 Active does not attack; Node 3 itself is not a legal attack target.
- **Declare Blocks:** Opponent blocks the Core attack with their Active, blocks the Node 1 attack with their Node 3 Active
- **Resolve Damage:**

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    - Core combat: Both Actives deal damage simultaneously, one is destroyed
    - Node 1 combat: Your Active survives, enemy Node 3 Active destroyed
    - Node 3 becomes neutral after combat: its enemy Active was destroyed and you have no Active there. Your Node 1 attacker remains in Node 1.

Phase 5: End Phase

- Check destroyed cards → move to discard
- Node 3 remains neutral. On a later Main Phase, deploying an Active into its empty friendly slot would claim it if the opponent still has no Active there. This turn you retain only Nodes 1 and 2.
- Buff durations decrement

Phase 6: Cleanup

- Expire "this turn" effects
- Pass turn to opponent

2.2. OBJECTIVES

The primary objective of Overlink is to achieve victory by strategically controlling the battlefield while managing cards, resources, and Active-Bench formations. Players must decide when to attack, defend, reinforce their positions, contest Nodes, or focus their efforts on the shared Core.

The gameplay aims to encourage strategic planning, adaptation, resource management, and effective card combinations. Players are encouraged to construct decks that support their preferred playstyle and adapt their strategy according to the opponent's actions and the current state of the battlefield.

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Victory can be achieved through three distinct paths:

- Node Domination: Control 3 out of 4 Nodes
- Core Destruction: Reduce opponent's Core durability to 0
- Deck-Out: Opponent cannot draw a card during their Draw phase and has no cards in hand

2.3. STORY OUTLINE

Beginning

The story takes place in a future where the Overlink network has been developed as a system for connecting and coordinating advanced technological and academic resources. Different factions emerge around competing visions for how the network should be used and developed.

Rising Action

As the Overlink network expands, conflicts between the factions intensify. Each faction attempts to gain control over important Nodes and establish influence over the central Core. Commanders are recruited to lead their faction's cards and compete for control of the network.

Climax

The conflict reaches its peak as the factions compete for control of the Core and the remaining Nodes. Every commander must use their available resources, formations, and strategies to overcome their opponent and secure their faction's position within the Overlink network.

Ending

The outcome of each conflict determines which faction gains control of the network's key areas. The Overlink system remains active, leaving

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the larger struggle between the factions unresolved and allowing new battles and rivalries to emerge.

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3. CHARACTERS

Player Commander

The Architect

Personality: Strategic, adaptable, and analytical.

Role: Represents the player during matches and commands their selected faction and card deck.

AI Commander

The Operator

Personality: Calculating, methodical, and competitive.

Role: Represents the AI opponent in single-player matches and makes strategic decisions based on the current battlefield.

Multiplayer Commander

The Challenger

Personality: Competitive and unpredictable.

Role: Reserved for a future online multiplayer mode; direct online matches are deferred beyond MVP. The Challenger represents another player's chosen deck and strategy if that mode is approved later.

4. SYSTEM DESIGN

4.1. ARCHITECTURE OVERVIEW

Overlink uses a three-layer architecture with dependencies pointing strictly inward. The Core layer contains all game rules and has no knowledge of Unity, rendering, or the network. The Data layer holds card and configuration definitions. The Presentation layer is the only layer that touches UnityEngine.

This inversion is the load-bearing decision of the project. Because Core has zero UnityEngine references, it compiles as a plain .NET class library that loads unmodified in the client, in a headless dedicated server, and in the NUnit test project. The same rules

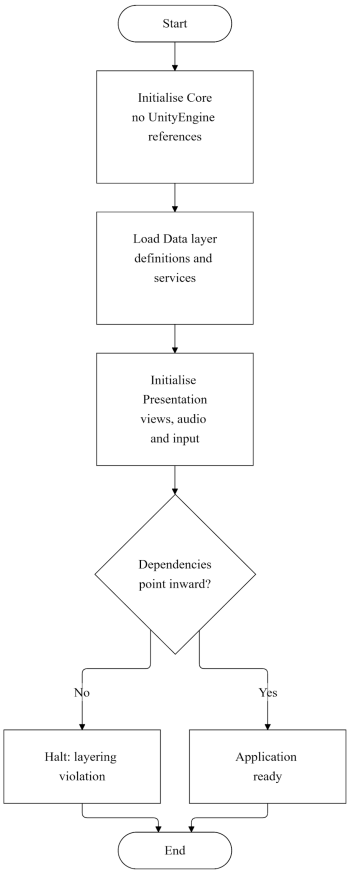
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binary enforces every rule on the server and drives every local simulation, so the two can never diverge.

Layer	Contents	Depends on
Core	GameState, phase machine, NP economy, combat resolution, victory checks, effect interpreter	.NET BCL only
Data	Card definitions, effect descriptors, progression and economy config	Core structs
Presentation	Views, animation, audio, UI, input	Core and Data

Online services are consumed through a narrow interface owned by the Data layer, so that Relay, Cloud Save, and Economy remain replaceable and testable. No Core module references a Unity service SDK directly.

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4.2. MODULE BREAKDOWN

Module	**Responsibility**	**Layer**
GameState	All mutable match state: slots, zones, counters, turn number	Core
PhaseMachine	Advances the six phases defined in 2.1.2 and guards legal transitions	Core
NPEconomy	NP gain, cap enforcement, cost	Core

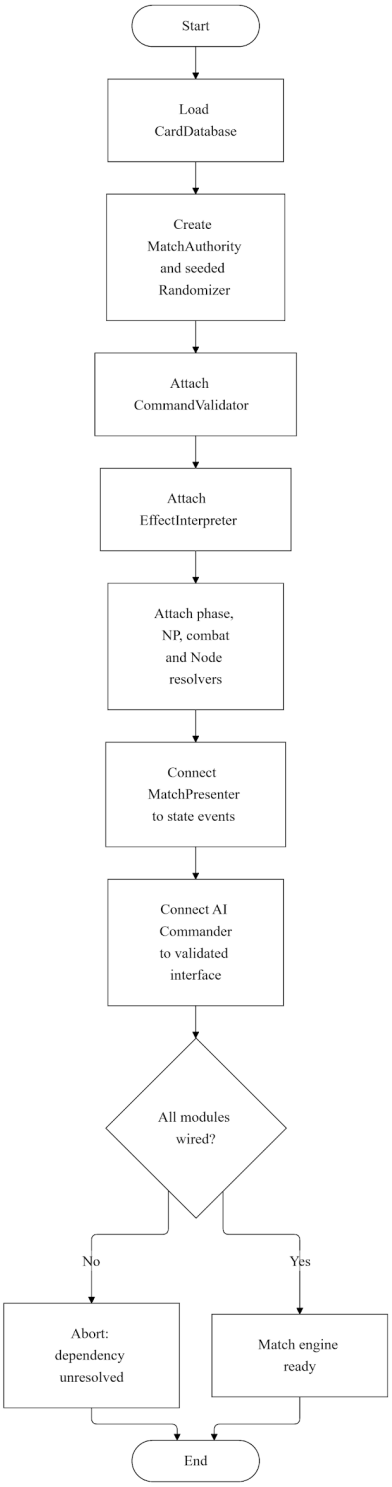
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	payment, persistence	
CombatResolver	Attack declaration, blocking, damage assignment, destruction	Core
NodeController	Node states, Core durability, Core multipliers	Core
EffectInterpreter	Executes effect descriptors emitted by cards	Core
CardDatabase	Loads and validates every card definition at startup	Core
CardInstance	One card in one zone, with mutable runtime state	Core
Randomizer	Seeded PRNG; the only randomness source in Core	Core
CommandValidator	Rejects illegal commands with player-facing reasons	Core
MatchAuthority	Owns authoritative state and resolves submitted commands	Core
ReplayRecorder	Records seed and command log for verification	Core
ServiceGateway	Interface wrapping Authentication, Relay, Lobby, Cloud Save, Economy	Data
SaveMapper	Translates Core state to	Data

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	and from Cloud Save schema	
EventChannel	ScriptableObject carrying UnityEvents for presentation	Presentation
MatchPresenter	Renders GameState as board visuals	Presentation

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4.3. CORE RULES ENGINE

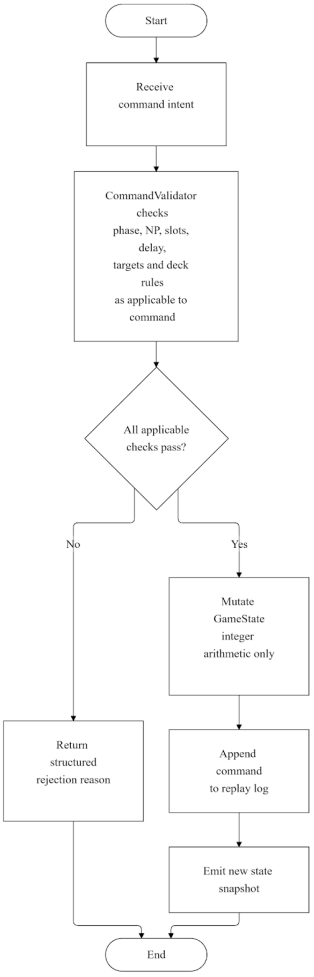
All match state lives in a single plain C# object, GameState. Commands are the only way to mutate it. The engine accepts a command, validates it against current state, and returns either a new state or a structured rejection reason. Nothing mutates state directly.

Commands represent player intent rather than results: DeployCard, SwitchPosition, DeclareAttack, DeclareBlock, ActivateAbility, and Concede. Keeping intent separate from outcome is what allows the server to remain authoritative, and makes every state change traceable in the replay log.

Validation is centralised in CommandValidator and covers phase legality, NP cost, slot availability, Summoning Delay, target legality, and deck construction rules. Because the AI and the human client call the same validator, neither can produce an illegal action.

The engine uses integer arithmetic only. Floating-point values are banned in Core, because rounding differences between the client and the server would produce divergent states that are expensive to diagnose.

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4.4. CARD AND ABILITY SYSTEM

Abilities are data, not code. A card holds an ordered list of effect descriptors, each an effect type with typed parameters. The EffectInterpreter is a single switch over effect types, so adding a new card requires no new C# beyond an existing effect type.

UnityEvent-based abilities were considered and rejected. Effects must serialise into save data and network messages, which UnityEvent cannot do. Inspector-authored code bypasses validation and could grant illegal values such as arbitrary NP or direct card movement. Headless NUnit tests require no scene setup, whereas delegate-driven effects leak when listeners are not explicitly removed. The effect vocabulary is defined by the EffectType enum in 4.15.2: DealDamage, DealAoEDamage, Heal, Draw, GainNP, Buff, Debuff, GrantShield, Destroy, Summon, Move, Exhaust, AddKeyword, SwitchPosition, PreventBlock, and Repair. These operations implement the card keywords and effects described in 2.1.11. 4.15.2 is the authoritative definition of this enum, and this list, the authoring dropdown, and the runtime switch must match it exactly.

UnityEvents remain the correct tool in Presentation, where wiring is transient, local, and never validated. This is the division the rest of the document follows.

A data-oriented ECS was also considered and rejected. The mutable board is on the order of one hundred entities changing a few times per turn, which is a latency problem rather than a throughput problem, and ECS would add a debugging tax without a matching performance gain.

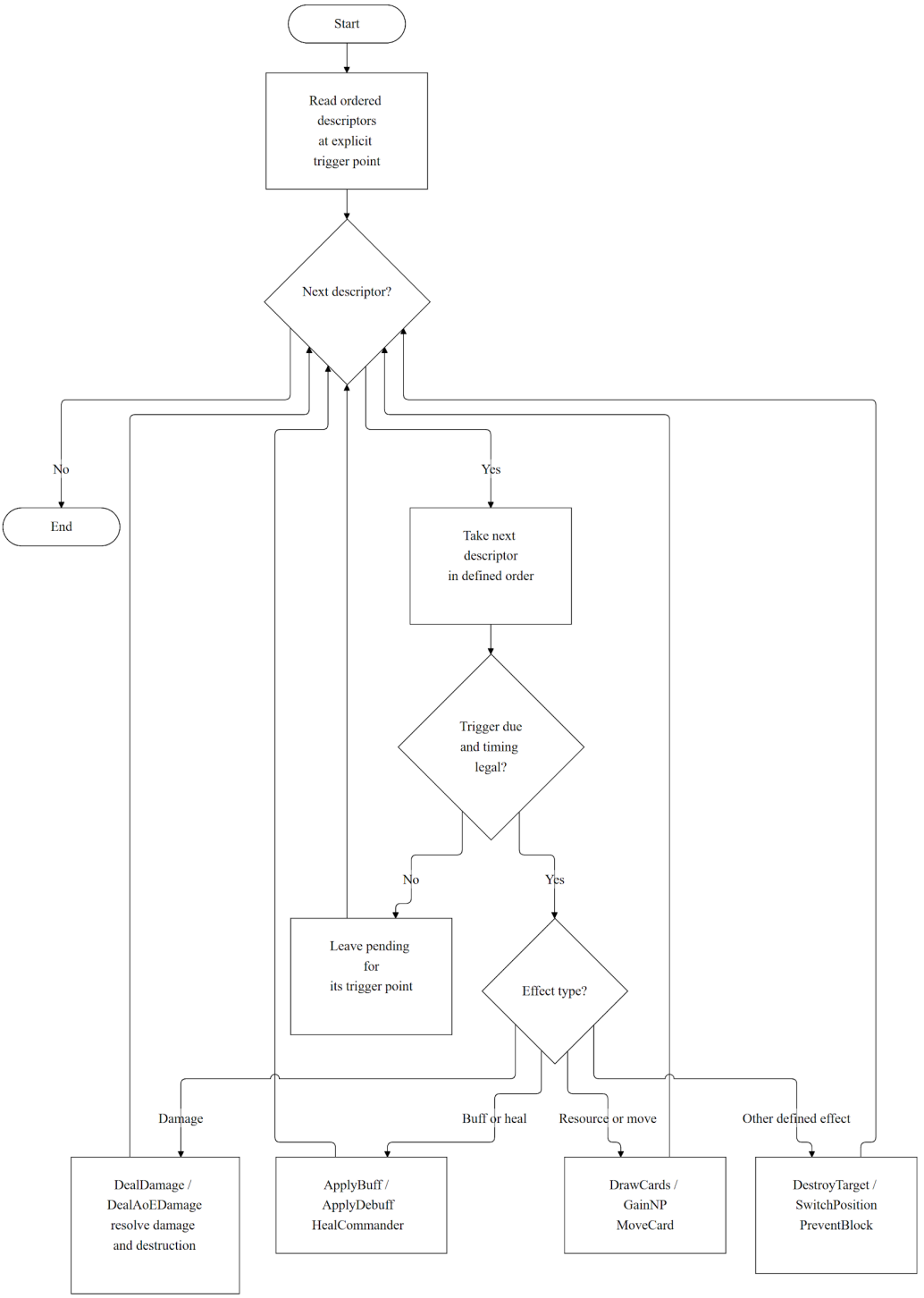
Effects may carry the [Fast] tag. Only the non-active player may use a [Fast] card/ability in the opponent's Main Phase between resolved commands, at most one per command; it resolves immediately and cannot trigger a nested response. [Fast] does not

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create windows in other phases or during combat resolution. Triggers resolve at explicit subscription points in Upkeep, Draw, Main,

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End, and Cleanup, so resolution order is deterministic.



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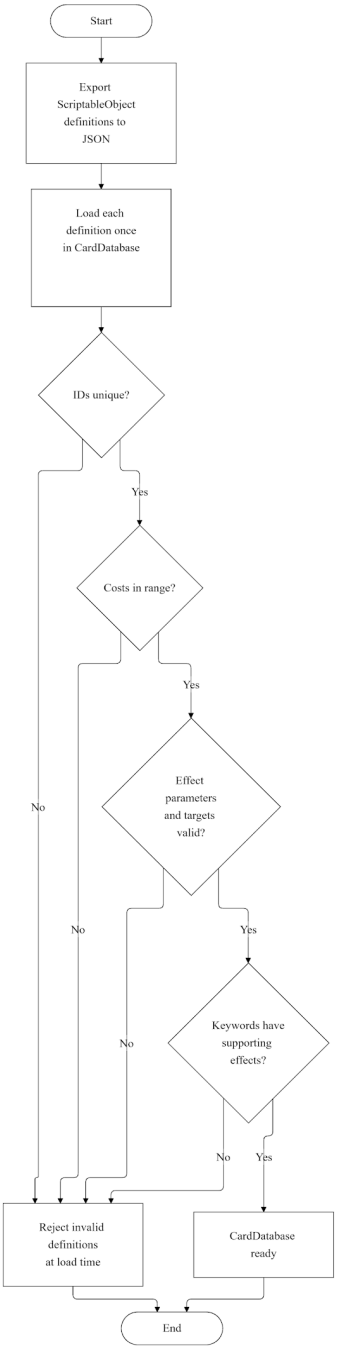
4.5. DATA LAYER

Card definitions are authored as ScriptableObjects in the editor and exported to JSON for runtime loading. Each definition holds identity, cost, a Type of Unit, Ability, Relay, or Protocol, stats, keywords, effects, rarity, and faction. Relays are placed in a Bench slot; Protocols are placed in the Protocol slot. Type names are serialized strings and may only be appended to the enum, never reordered or reused.

Card identity is a stable string key that never changes across releases. Players own randomised collections, so renumbering or reordering the pool would silently corrupt existing accounts. This constraint is recorded in Risk Assessment.

CardDatabase loads every definition once and validates the pool at startup: no duplicate IDs, costs within range, no effect referencing a missing target, and no keyword without a supporting effect. Invalid definitions are rejected at load rather than failing mid-match.

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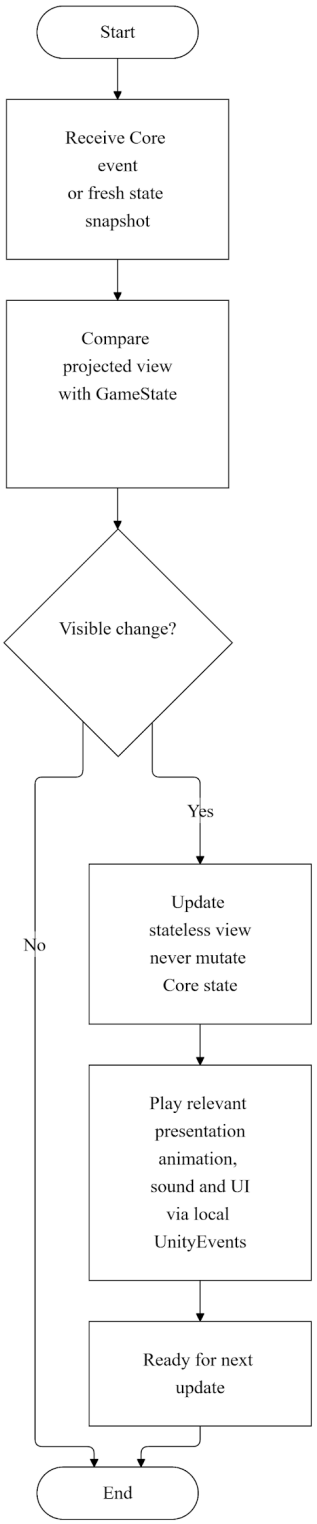


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4.6. PRESENTATION LAYER

Presentation subscribes to Core events and never mutates state. A card entering a zone triggers an entry animation, sound, and UI highlight; a card taking damage triggers a damage popup and screen shake. Views are stateless projections of GameState and can be rebuilt at any time, which is what allows a reconnecting client to recover from a fresh snapshot alone.

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4.7. NETWORKING ARCHITECTURE

MVP matches are single-player and resolved by the local Core rules engine. Account sign-in, progression saves, and pack opening require an active online service connection; MVP does not support offline accounts; anonymous guest accounts are supported. Online account/cloud-save and pack-opening services validate their own requests server-side. Direct online matches and authoritative multiplayer match state are deferred beyond MVP.

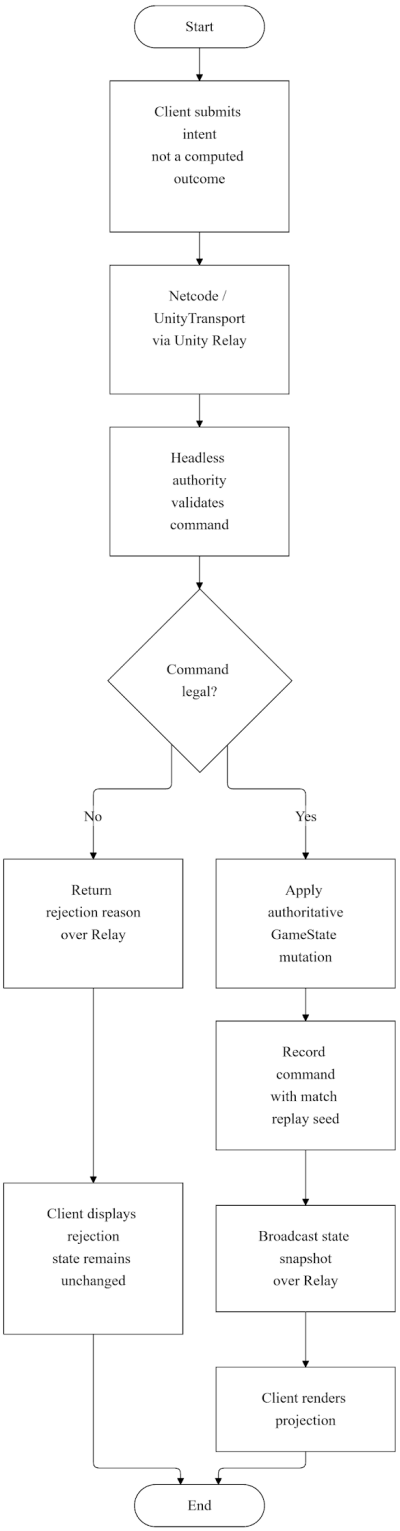
Post-MVP multiplayer architecture candidate: Unity Relay would carry traffic for WebGL clients; Netcode for GameObjects and UnityTransport would provide session/command plumbing, and Lobby/Matchmaker would group players. None of Relay, NGO multiplayer sessions, Lobby, or Matchmaker is a shipping MVP requirement.

For a future online mode, the proposed model is strict command-and-response because cards are face-up and matches are turn-based. Prediction, reconciliation, and interpolation are not MVP requirements and remain subject to a post-MVP multiplayer prototype.

In MVP, the local match engine owns a seeded PRNG and records the seed and command log for deterministic local replay/testing. For a future online mode, the server would own the PRNG and authoritative command log; multiplayer replay service is deferred.

A dedicated headless authoritative match server is a post-MVP design option, not part of the first public release. MVP ships local single-player matches; the Unity-independent Core assembly is shared by the client and test project. If online matches are approved later, a server build and deployment become a separately estimated milestone.

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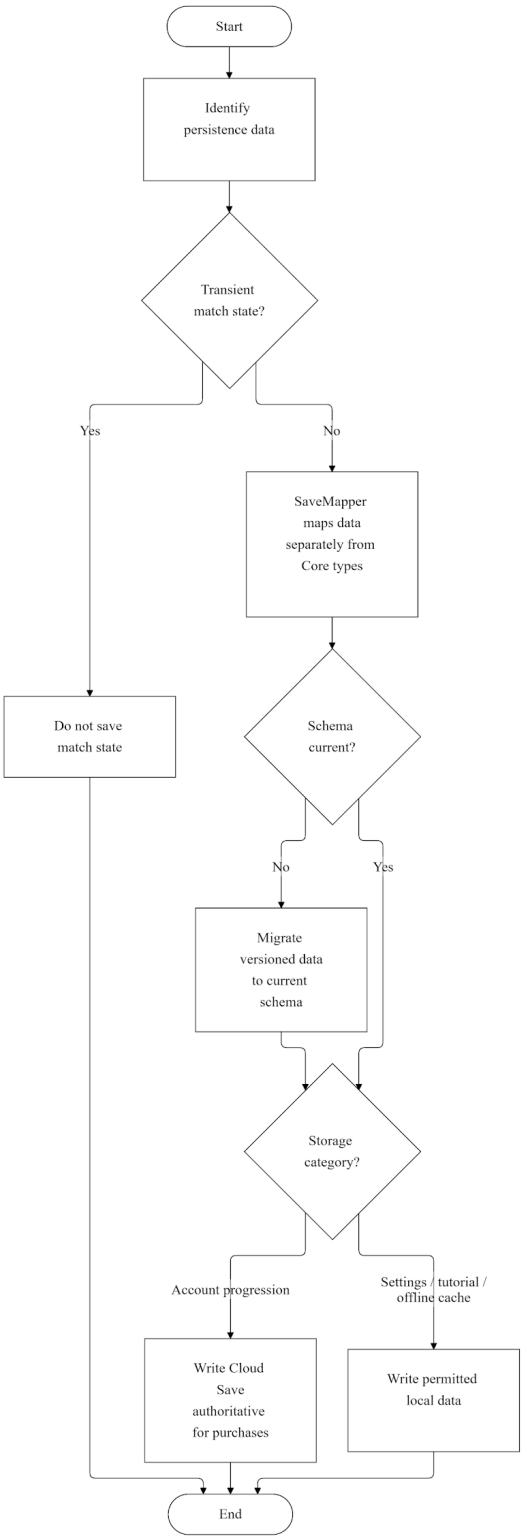
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4.8. SAVE SYSTEM

Match state is transient and never saved. Progression state, including unlocked card IDs, deck definitions, and settings, is stored through Cloud Save.

Save data carries a schema version so future changes migrate rather than discard. Cloud Save is the authoritative store for all authenticated progression, including unlocked card IDs, deck definitions, tutorial progress, and settings. MVP does not support offline progression or an offline account; local storage may retain non-authoritative presentation preferences only. The SaveMapper keeps the Cloud Save schema separate from Core types so the wire format can evolve without touching the rules engine.

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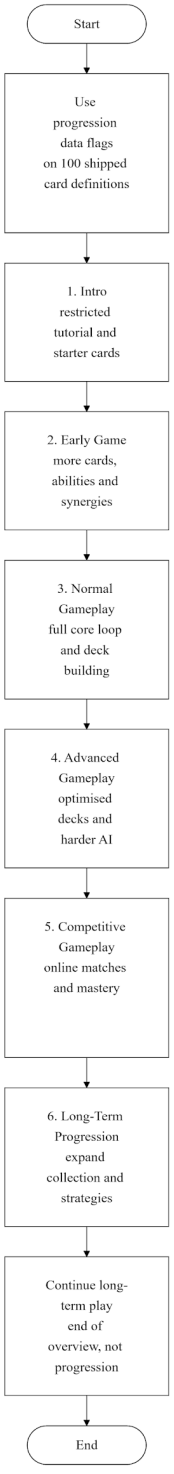


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4.9. PROGRESSION SYSTEM

Progression follows the six stages in Section 6. All 100 card definitions are shipped as data as specified in 2.1.9 and 4.15.6; the cards available to a player at each stage are controlled by progression flags on those definitions. The tutorial uses a restricted rules configuration with a starter subset of cards and exposes only deployment, combat, NP, and Node control. Later stages enable additional card pools and ability types through data flags, so progression is content configuration rather than branching code.

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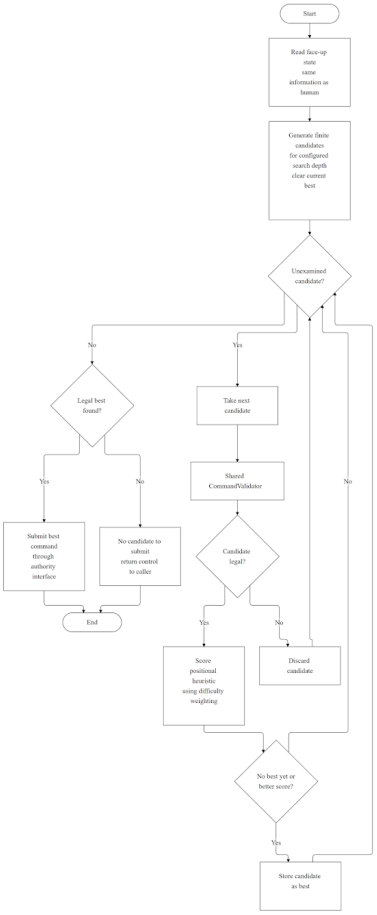
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4.10. AI SYSTEM

The AI Commander issues commands through the same validated interface a human player uses, and is subject to the same legality checks. Because all cards are face-up, the AI reads exactly the information a player sees and needs no information-hiding heuristics.

The AI scores candidate commands using positional heuristics rather than deep search, which suits a solo developer and keeps match time short. Difficulty levels adjust search depth and evaluation weighting only, not the rule set.

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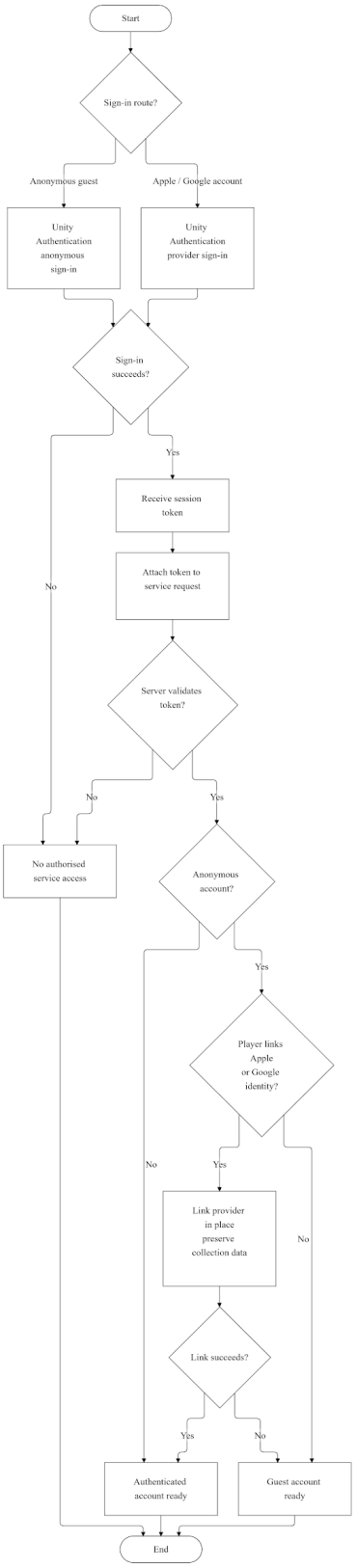
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4.11. AUTHENTICATION

Authentication uses Unity Authentication with Google or another supported identity provider. A player must complete online sign-in before entering the game; MVP does not provide offline accounts or offline progression. Sign-in may be anonymous, creating a guest account, or linked to a supported identity provider. No bespoke password system is built, so the project never stores credentials.

A successful sign-in yields a session token validated on every service request. The authority derives all authoritative state from that token, so a client cannot prove ownership of cards it does not hold. A returning Apple or Google account proceeds directly as an authenticated account. Only anonymous guest accounts are offered an upgrade path; linking an identity provider upgrades the guest in place and preserves its collection data.

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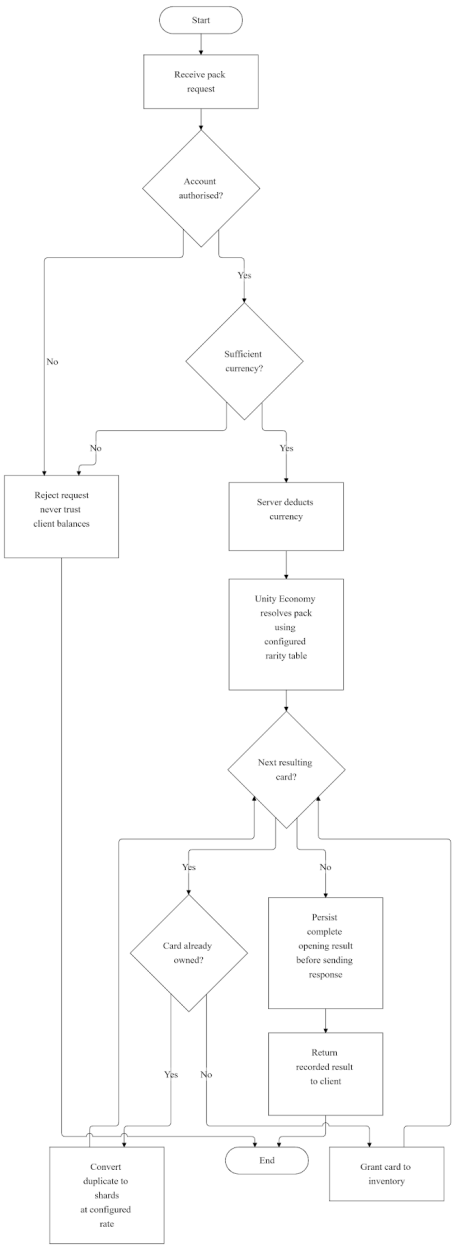
4.12. MONETIZATION AND ECONOMY

Overlink MVP uses free-to-play single-player progression and earned in-game currency. Real-money purchases and paid currency are deferred; online pack-opening requests, when available, are validated server-side. See 10.1 and 12.

When online, Unity Economy resolves each 5-card pack server-side using the per-slot odds in 2.1.9, deducts earned in-game currency, and returns the resulting card IDs. The opening and currency deduction are recorded atomically before the response is sent, so a dropped connection cannot lose currency or duplicate a pack. Duplicate cards convert to shards at the configured rate. Offline play does not permit pack opening; MVP has no real-money purchases or paid currency.

Real-money payments, paid currency, and payment-provider integration are deferred beyond MVP. No payment or purchase flow ships in the first public release.

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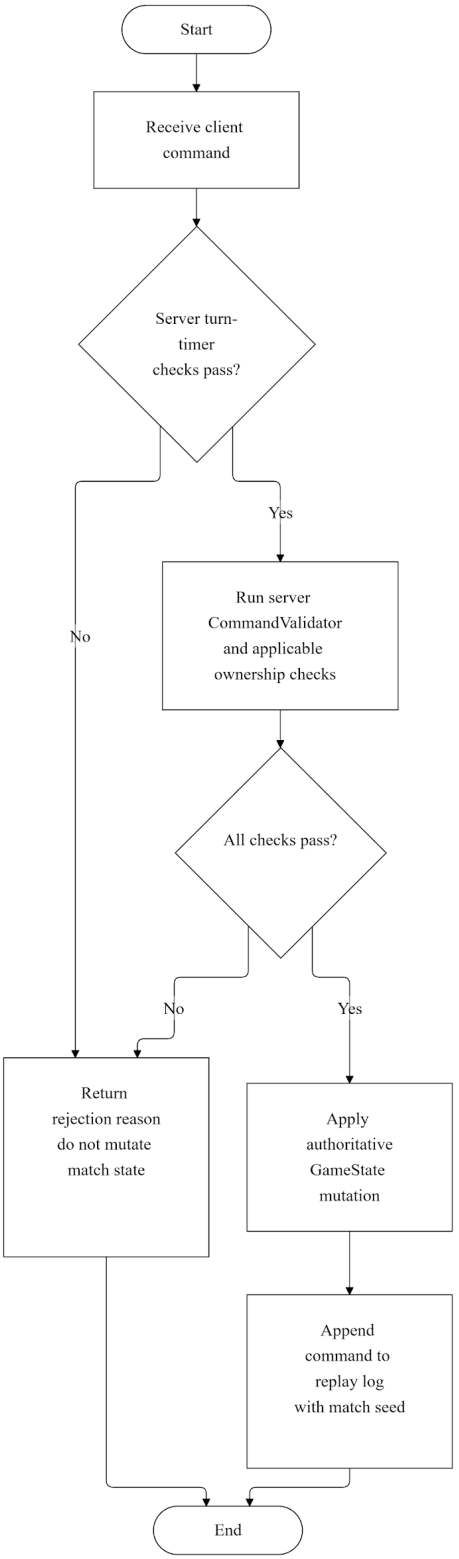
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4.13. ANTI-CHEAT

Anti-cheat is a consequence of the architecture rather than a separate system. Because the authority validates every command and owns all progression state, a modified client can at most render incorrect visuals, which it already controls locally.

Two residual risks remain. A client can submit commands faster than a human, so turn timers are validated server-side. And a client can display forged state to itself, which affects only its own play experience and is acceptable for a competitive-but-casual title.

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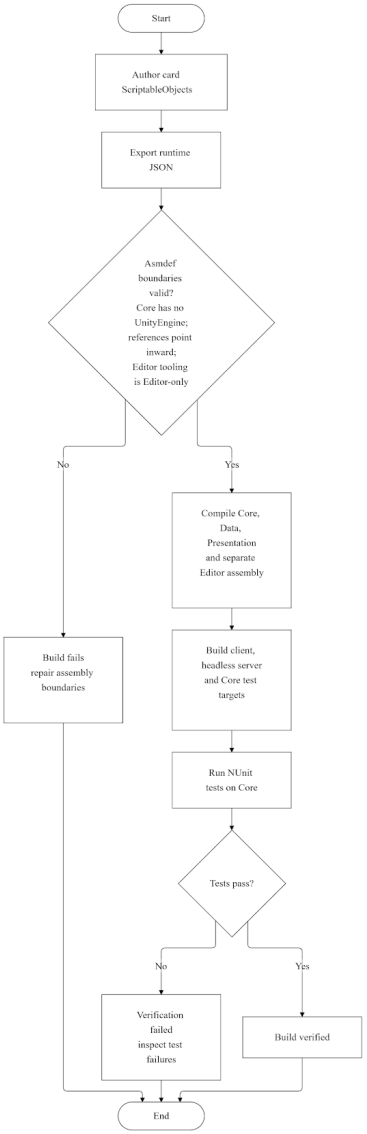
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4.14. ASSEMBLY DEFINITIONS

Code is split into asmdef boundaries matching the layer boundaries, which makes the inward dependency rule a compile-time error rather than a review convention.

Core compiles as a plain .NET standard library referenced by the client, the dedicated server project, and the test assembly. Data references Core. Presentation references both. Editor tooling sits in a separate asmdef restricted to the Editor platform, keeping it out of player builds.

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4.15. EDITOR TOOLING

Card content is authored through editor tooling rather than through code. A designer creates a card asset, fills the fields defined in 2.1.10, and exports; the runtime never sees a hardcoded card. Adding a card to the pool is routine design work and requires no programming.

This section specifies the tooling, the rules it enforces, and the workflow that produces the runtime data described in 4.5.

4.15.1. CARD DEFINITION ASSET

Each card is a single `ScriptableObject` asset in the editor, created via **Assets > Create > Overlink > Card Definition**. Its fields follow the anatomy template in 2.1.10, extended with the identity, rarity, and faction fields required by 4.5.

Field	Type	Inspector	Notes
ID	String	Text field	Stable key that never changes across releases. Hand-authored, not generated. Format: `card_<faction>_<nnn>`
Name	String	Text field	Player-facing card name
Faction	Enum	Dropdown	One faction per card; MVP decks are single-faction per 2.1.9
Rarity	Enum	Dropdown	Common, Rare, Epic, or Legendary
Type	Enum	Dropdown	Unit, Ability, Relay,

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			or Protocol
Cost	Integer [0–10]	Int slider	NP required to deploy; ceiling of 10 from 2.1.1
ATK	Integer ≥ 0	Int field	Damage dealt in combat
HP	Integer ≥ 0	Int field	Health before destruction
Shield	Integer ≥ 0	Int field	Damage absorbed before HP
Keywords	List<Keyword>	Reorderable list with dropdown per row	Drawn from the MVP set in 2.1.11
Ability Text	String	Multi-line text area	Player-facing rules text, matched to the effects below
Design Note	String	Multi-line text area	Internal rationale; exported for tooling, excluded from player builds
Effects	List<EffectEntry>	Reorderable list with foldout per effect	Ordered effect descriptors, per 4.4
Art Path	String	Text field	Location of the card illustration

Required fields (ID, Name, Faction, Cost) are enforced at author time via `OnValidate` rather than at runtime. A card with no ID, name, or faction is rejected before export rather than failing later during a play session.

4.15.2. EFFECT AUTHORING

Abilities are data. Each effect is a row describing what happens, to what, and by how much, matching the effect descriptors required by 4.4. Relays and Protocols are

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persistent card types: their printed effects apply only while the card remains in its defined slot.

Property	Type	Inspector	Purpose
Effect Type	Enum	Dropdown	Which operation resolves: DealDamage, DealAoEDamage, Heal, Draw, GainNP, Buff, Debuff, GrantShield, Destroy, Summon, Move, Exhaust, AddKeyword, SwitchPosition, PreventBlock, Repair. This is the authoritative EffectType enum and matches the vocabulary stated in 4.4. CaptureNode is not an authorable MVP operation: Node control is derived from Active occupation (2.1.7). If an enum value already exists, reserve it without reordering other values and reject it during validation.
Target	Enum	Dropdown	What the effect points at: Self,

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			Commander, EnemyCommander, AlliedActive, EnemyActive, AllAlliedActive, AllEnemyActive, Node, EnemyNode, Deck, Discard, Hand. Self, Commander, and EnemyCommander all address the player rather than a card: Self and Commander both refer to the controlling player, and EnemyCommander refers to the opponent's player. They have no ATK, HP, or Shield, and the only health pool a player owns is Core durability (2.1.7). Effects that act on a player resource (for example Draw, GainNP, or Repair) may target them. Any effect that requires a card to resolve (for example DealDamage, DealAoEDamage, Heal, or Destroy) is illegal against Self,

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			Commander, or EnemyCommander and must be rejected by validation. Repair restores Core durability, so it targets Self or Commander and is legal.
Amount	Integer	Int field	Primary magnitude for the effect
Amount 2	Integer	Int field	Optional secondary magnitude
Duration	Integer	Int field	Turns the effect lasts; instant when 0, permanent when -1
Reference ID	String	Text field	Card identity for effects that summon, move, or grant another card
Fast	Boolean	Toggle	Whether the effect may resolve during the opponent's turn, per 4.5

The current effect vocabulary covers damage, area damage, healing, card draw, NP gain, buffs and debuffs, shield, destruction, summoning, movement, exhaust, keyword granting, position switching, and block prevention. Operation names are canonical: Buff and Debuff (not ApplyBuff or ApplyDebuff), Draw (not DrawCards), Move (not MoveCard), and Destroy (not DestroyTarget). Summoning, movement,

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and destruction recompute Node control from occupation under 2.1.7; no effect directly assigns Node ownership or bypasses the occupation requirement.

A new card built from existing effects requires no code change. A genuinely new effect type is a deliberate code change to the EffectInterpreter described in 4.4, and is reviewed separately so that the data vocabulary stays deliberate rather than growing by accident.

4.15.3. CARD MANAGER WINDOW

A single editor window (**Overlink > Cards > Card Manager**) is the entry point for all card work. Designers do not need to locate assets manually or remember export paths.

Function	**Behaviour**
Pool browser	Lists every card asset with ID, rarity, type, and name; sorted by ID for stable review diffs
Filter	Narrows the pool by ID, name, faction, rarity, or type (text search + enum dropdowns)
Card inspector	Shows the selected card's key fields and links to its asset for editing
Validation display	Errors and warnings appear inline; cards with errors are highlighted in the list
Composition summary	Live counts per rarity and per faction
Export control	Writes the runtime data file and reports the result

4.15.4. VALIDATION RULES

Type-specific checks: the serialized Type is exactly Unit, Ability, Relay, or Protocol. Units may occupy only Active or Bench positions; Abilities resolve without a board slot; Relays may occupy only a Bench slot; Protocols use the single player-owned Protocol slot. Reject illegal type/zone pairs, non-Unit combat stats, a second

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simultaneously active Protocol without replacement, and effects whose target or scope is illegal for the card type. A persistent effect must end when its source card leaves its slot.

Validation runs in three places with identical rules: OnValidate for instant feedback, the Card Manager for batch review, and runtime startup (CardDatabase.Load). Both editor passes use the same CardValidator class, so a card that could never work cannot reach a build. This implements the pool validation required by 4.5.

Rule	**Severity**	**Where Checked**	**Rationale**
ID is present and unique across the pool	Error	Manager, Export, Runtime	Duplicate or missing IDs break the stable identity required by 4.5
Name and faction are present	Error	OnValidate, Manager, Export	A card with no identity cannot appear in a collection or deck
Cost is between 0 and 10	Error	OnValidate, Manager, Export	Above the 10 NP ceiling from 2.1.1 the card can never be played
Stats (ATK, HP, Shield) are not negative	Error	OnValidate, Manager, Export	Negative stats have no defined behaviour
Active Units have ATK, HP, or at least one effect	Error	Manager, Export	An Active Unit with nothing to do cannot participate in combat
Effects reference an existing card ID	Error	Manager, Export, Runtime	A dangling reference cannot resolve at play time
Effects that require	Error	Manager, Export	Whether this effect

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a magnitude have a positive Amount			may be used by the non-active player during the opponent Main Phase between resolved commands, as defined in 2.1.11
Keywords have a supporting stat or effect	Error	Manager, Export, Runtime	Prevents keywords that read as rules but resolve to nothing (4.5)
Keywords apply to the card type	Warning	Manager, Export	A keyword with no meaning on that card type is usually a mistake
Non-combat types (Benched Units, Abilities, Relays, and Protocols) carry no combat stats	Warning	Manager, Export	Combat stats on a Unit are read only while it is in an Active position; while a Unit is Benched, its stat values are read as buff deltas granted to its battlefield's Active Unit. Relays and Protocols have no combat stats and never occupy an Active position.
Card has ability text when it has mechanics	Warning	Manager, Export	Players cannot see a mechanic that is never described
Pool matches the rarity budget	Warning	Manager (when enforced), Export	Enforced only when the pool is

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(50/30/15/5)		(when enforced)	declared complete
Art is assigned	Info	Manager	Tracked during art production, not blocking
Self, Commander, or EnemyCommander targets require no card	Error	Manager, Export, Runtime	Self, Commander, and EnemyCommander address the player, not a card: they have no ATK, HP, or Shield, and the only player health pool is Core durability (2.1.7). Effects that need a card to resolve must be rejected; Repair targets Core durability and is legal (4.15.2).
Repair targets the player and is capped at the starting Core durability	Error	Manager, Export, Runtime	Repair restores Core durability to Self or Commander and cannot exceed the starting value of 3; Repair against any other target is illegal (2.1.7, 4.15.2).

Errors block export. Warnings are reported and do not block, because a pool under construction is expected to be temporarily incomplete.

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4.15.5. EXPORT PIPELINE

Export produces the runtime data file described in 4.5. The card assets are the source of truth, and the exported file is a build artefact regenerated from them.

Stage	**Behaviour**
Collect	Every `CardDefinitionSO` in the project is gathered via `AssetDatabase.FindAssets`, loaded, and ordered by ID for reviewable diffs
Validate	The full rule set runs via `CardValidator`; any error halts the export
Serialise	Valid definitions are written as JSON with enums recorded by name (not ordinal) using a hand-rolled codec in `CardJson` (Core assembly has no UnityEngine dependency)
Import	The output file (`Assets/Overlink/Generated/cards.json`) is registered via `AssetDatabase.ImportAsset` so it is visible and version-controlled
Report	Card count, warning count, and the output path are logged to the Console

Enums are written as names rather than ordinals so that reordering an internal enum cannot corrupt existing card data. The exported file carries a format version, and a build refuses to load data declaring a newer version than it understands rather than silently dropping fields.

The export is invoked via **Overlink > Cards > Export Card JSON** (Ctrl+Shift+E) or from the Card Manager window.

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4.15.6. SEEDING THE MVP POOL

The MVP library's four card types and exact rarity counts are defined in 2.1.9. These collection shares describe the catalog, not pack odds; packs use the independent per-slot odds specified there.

Composition is balanced against the reference deck rather than against percentages that do not resolve to a card count. The 30-card reference deck of 16 Units, 6 Abilities, 4 Relays, and 4 Protocols scales to the full pool, and each new faction is balanced against that distribution.

Order	**Work**	**Notes**
1	Draft card lists per faction on paper or in a spreadsheet	Text is enough at this stage; ID, cost, type, rough mechanics
2	Create card assets in the editor	Bulk work, no code; use **Create > Overlink > Card Definition**
3	Validate after each batch	Errors surface while the design is still fresh; use Card Manager
4	Assign art and final ability text	Art production runs in parallel, per 7.3
5	Export and boot the game against the exported file	Confirms the runtime path end to end

4.15.7. VERSION CONTROL POLICY

Card identity is treated as frozen data. Because player collections reference cards by ID, changing an ID after release silently corrupts existing accounts, as noted in 4.5.

Rule	**Reason**
Card IDs are never renumbered or	Player collections reference them

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reused	permanently
New cards receive new IDs appended in sequence	Keeps history readable
The exported `cards.json` is committed alongside the assets	Reviewable diffs and reproducible builds
Balance changes edit existing cards rather than replacing them	Preserves identity across balance passes
Bulk find-and-replace across card IDs is prohibited	No tooling can detect the account damage it causes

4.15.8. TECHNICAL NOTES FOR IMPLEMENTATION

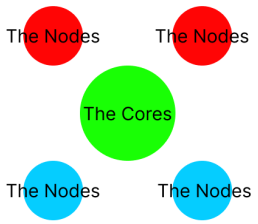
Concern	**Decision**
Effect data structure	Flat `EffectDescriptor` struct (not `SerializeReference` polymorphism). Simpler JSON, stable switch in EffectInterpreter, sufficient for MVP keyword set.
Keyword list in inspector	`ReorderableList` with per-row dropdown; Keywords are not bitwise flags.
Enum stability	Enums in Core (`CardType`, `Rarity`, `Keyword`, `EffectType`, `TargetKind`) must never reorder or remove values. New values append only.
Cross-asset references	Reference ID strings validated at export and runtime; no `Object` references across assets (breaks when assets move).
Generated folder	`Assets/Overlink/Generated/` excluded from `.gitignore` so `cards.json` is

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	committed.
Build pipeline	`CardJsonExporter.Export` can be called from CI via `-executeMethod` for automated validation.

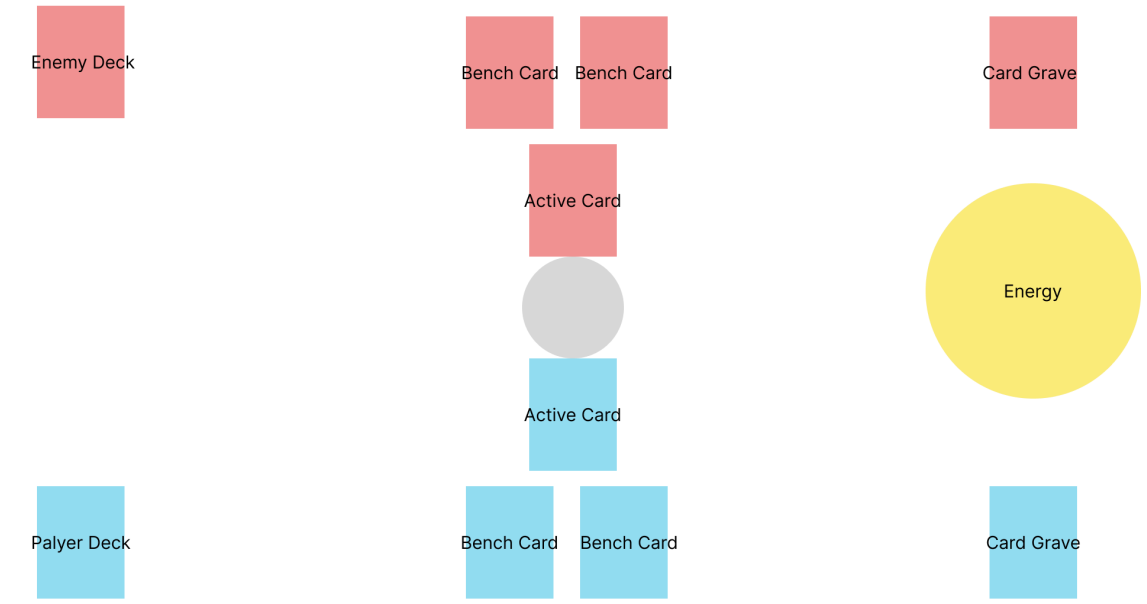
5. LEVEL DESIGN

The following figures are spatial references for the Node and Core play areas, not additional rules. They illustrate the Nodes layout and the Nodes/Core arena arrangements; slot counts, ownership and control, deployment eligibility, and combat are governed by 2.1.3–2.1.7. Do not infer extra gameplay rules from diagram spacing or decoration.

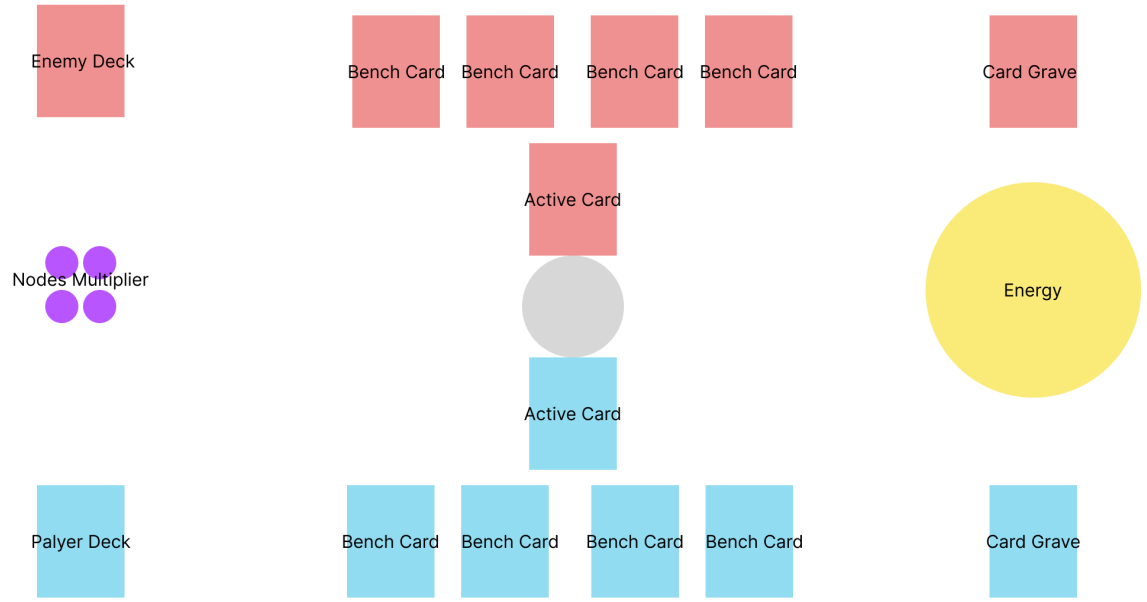


Battle Arena Layout (Nodes)

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Battle Arena Layout (Cores)



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6. PROGRESSION

Intro

The player is introduced to the basic rules through a tutorial match. Only the essential mechanics are available, including card deployment, Active and Bench positions, basic combat, resource usage, and Node control. The tutorial guides the player through a simplified match to establish the fundamental gameplay loop.

Early Game

The player gains access to additional cards and begins experimenting with different card combinations. More card abilities, Bench synergies, and basic deck-building options become available. Matches gradually introduce more complex decisions involving Node control and resource management.

Normal Gameplay

The full core gameplay loop becomes available. Players can build and customize decks, collect cards through gameplay and card packs, and experiment with different faction strategies. All major battlefield mechanics, including the Core, four Nodes, Active-Bench interactions, and card synergies, are available.

Advanced Gameplay

Players begin optimizing their decks and developing more specialized strategies. Higher-rarity cards, stronger card combinations, and more complex interactions encourage players to understand the strengths and weaknesses of their decks. The AI opponent also provides more challenging matches by making more strategic decisions.

Competitive Gameplay

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Direct online matches are deferred beyond MVP and are not required for this progression stage. If approved for a later milestone, competitive progression would focus on deck composition, matchup knowledge, and battlefield control.

Long-Term Progression

Players continue expanding their card collection, experimenting with different decks, and improving their strategic understanding. New cards and card combinations provide additional possibilities without requiring players to relearn the fundamental gameplay system.

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7. ARTS AND VISUALS

7.1. ART DIRECTION

Overlink presents a near-future network contest: abstract technological architecture, contested signal space, and the people who fight over it. The visual target pairs Shadowverse's painterly, high-contrast card illustration with Ingress's geometric, scan-line aesthetic.

Shadowverse supplies the character layer. Cards are illustrated with readable silhouettes, dramatic lighting, and a single clear focal point, because the art must still work at the size of a Bench slot and at thumbnail size in a collection list.

Ingress supplies the world and system layer. Faction territory is read through luminous geometric overlays, scanning contours, and a dark ground that lets signal colour carry all the meaning. The same visual language expresses Node control on the battlefield.

Style Pillars

Signal over hardware. The Overlink network is shown as light, scanning lines, and data traces rather than as literal machines.

Painterly figures, geometric systems. Commanders and card characters are rendered with texture and depth; the network around them is rendered with clean geometry.

Readable without text. Every important battlefield state must be legible at a glance, before the player reads any label.

Restrained emission. Glow signals state and hierarchy and is never purely decorative, which also keeps the scene affordable on low-to-mid-range hardware.

Palette

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Backgrounds sit at the dark end of the value range, so card art and signal colour supply the contrast.
Each faction owns one signature hue, used for its territory, its card frame accent, and its commander lighting.
Neutral greys are reserved for frames, text, and controls, and are never used as faction colour.

7.2. VISUAL RULES

The following distinctions must be readable from the battlefield view alone.

Faction Identity

A faction is identified by hue plus silhouette motif, never by hue alone, so factions stay distinguishable for colour-blind players.
Card illustration stays within the faction palette; faction identity is carried by the frame rather than by repainting every illustration.

Battlefield State

Node control is the most important state to read. A controlled Node is marked by persistent illuminated geometry, and a contested Node shows active scan motion.
Cards under Summoning Delay are visually dimmed until the restriction clears, matching the rule in 2.1.5.
Durability is shown as wear on the card itself rather than as a separate numeric badge.
Damage, healing, and destruction each trigger a short high-contrast flash so the player never misses the outcome of an action.

Card Frame Hierarchy

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Rarity is expressed through frame complexity and ornament rather than colour alone: Common is plain, Rare adds a single accent, Epic adds a second accent and an inner border, and Legendary adds an animated outer treatment.
Because rarity spans the four tiers defined in 2.1.9, each frame tier must remain distinguishable in greyscale.

Animation Language

Entering a zone, leaving a zone, taking damage, healing, and destruction each receive one consistent animation shared across all factions.
Animation is presentation only and never affects game state, consistent with the layering rule in 4.6.
No state change that a player must react to runs longer than roughly half a second.

7.3. SPRITES REQUIREMENT

Asset locations are project-relative. Priority reflects what blocks the MVP.

Faction and Card Assets

Asset	Location	Priority
Faction 1 portrait - commander	Assets/Art/Factions/F1/ Commander	Critical
Faction 1 territory overlay	Assets/Art/Factions/F1/ Territory	Critical
Faction 2 portrait - commander	Assets/Art/Factions/F2/ Commander	Critical
Faction 2 territory overlay	Assets/Art/Factions/F2/ Territory	Critical
Faction 3 portrait -	Assets/Art/Factions/F3/	Critical

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commander	Commander	
Faction 3 territory overlay	Assets/Art/Factions/F3/Territory	Critical
Card frame - Common	Assets/Art/Frames/Common	High
Card frame - Rare	Assets/Art/Frames/Rare	High
Card frame - Epic	Assets/Art/Frames/Epic	High
Card frame - Legendary	Assets/Art/Frames/Legendary	High
Card art - Common set	Assets/Art/Cards/Common	High
Card art - Rare set	Assets/Art/Cards/Rare	Medium
Card art - Epic set	Assets/Art/Cards/Epic	Medium
Card art - Legendary set	Assets/Art/Cards/Legendary	Low
Card thumbnail - collection list	Assets/Art/UI/Thumb	High

Battlefield and Effects

Asset	**Location**	**Priority**
Battlefield background	Assets/Art/Field/Background	Critical
Node geometry - neutral	Assets/Art/Field/Node_Neutral	Critical
Node geometry - controlled	Assets/Art/Field/Node_Controlled	Critical

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Node geometry - contested	Assets/Art/Field/Node_Contested	High
Core geometry	Assets/Art/Field/Core	Critical
Active slot marker	Assets/Art/Field/ActiveSlot	High
Bench slot marker	Assets/Art/Field/BenchSlot	High
FX - card enters zone	Assets/Art/FX/ZoneEnter	Critical
FX - card leaves zone	Assets/Art/FX/ZoneLeave	Medium
FX - damage impact	Assets/Art/FX/Damage	Critical
FX - heal	Assets/Art/FX/Heal	Medium
FX - destruction	Assets/Art/FX/Destroy	Critical
FX - summon delay dim overlay	Assets/Art/FX/SummonDelay	Medium
Deck and discard pile art	Assets/Art/Field/Piles	Low

8. UI AND HUD

Overlink is a mouse-and-keyboard game released for PC and Web, so every screen below is designed in Figma first and then rebuilt in-engine against that file. The Figma file is the review artifact; the in-engine build is the source of truth for behaviour.

8.1. UI RENDERING BASIS

The runtime interface is built with uGUI (Canvas) and TextMeshPro. UI Toolkit was considered and rejected as the MVP runtime basis.

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Consideration	Decision
Web target	Canvas renders through one path on both PC and Web. UI Toolkit on WebGL introduces a second render path that has repeatedly produced layout and build regressions during prototyping
Team familiarity	Canvas with TextMeshPro matches the team's existing Unity work. UI Toolkit requires authoring UXML and USS as a parallel skill set for no MVP benefit
Visual composition	Most of the interface is composited sprites, rarity frames, and effect overlays rather than styled text, which Canvas handles predictably at high instance counts
Retained path	UI Toolkit remains viable for internal debug and developer tooling screens, which never ship to players

Layout is authored once at a 1920x1080 reference resolution using a Canvas Scaler in Scale With Screen Size mode, matching the 16:9 target in 1.4. Edge-anchored elements use anchors; repeated containers such as the hand and the bench use layout groups so slots reflow without manual repositioning.

8.2. FONT

Type is bundled and never resolved from the system. A machine or browser missing the chosen font would otherwise substitute silently, which reflows every label and breaks the layout the Figma file specifies. This matters most on Web, where the available font set differs per browser and platform.

Role	Typeface	Weight	Use
Title	Rubik	Bold	Screen titles, section headings in menus

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Body	Rubik	Regular	Body copy, descriptions, rules text
Numeric	Rubik	SemiBold	NP totals, ATK and HP values
Keyword	Rubik	Bold Italic	Keyword names such as [Piercing] in rules and card text

Rubik is licensed under the SIL Open Font License, which permits bundling in a distributed game without per-title licensing. Regular and bold are subset to the glyphs required by localised strings to limit Web download size. On Web the font loads from the build's own asset bundle rather than a CDN, so rendering is identical offline.

Stat values use tabular figures so digits do not shift horizontally as NP totals change during a turn.

8.3. MENU

Each screen is specified in Figma with its full state set, then implemented as a Canvas prefab.

Screen	**Contents**	**States**
Main Menu	Play, Deck Builder, Collection, Settings, Quit	Default, hover, pressed, focus ring, disabled during load
Settings	Resolution and window mode, master/music/SFX volume, fullscreen, language, account link	Default, unsaved-changes prompt
Pause	Resume, Settings, Abandon Match	Default, confirm-on-abandon

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Loading	Determinate progress where assets allow, spinner fallback	Progress, spinner, error with retry
Game Over	Result, Core durability remaining, Core cards lost, rematch, main menu	Victory, defeat, disconnect
Confirmation	Generic destructive-action modal for abandon-match and discard-deck	Default, focused confirm, focused cancel

Every menu is fully keyboard-operable as well as mouse-driven, consistent with 1.5. Focus is always visible, follows a predictable order, and never depends on the pointer. No menu state is reachable by pointer alone.

8.4. HUD

The battlefield HUD is a stateless projection of GameState and never mutates state, following the layering rule in 4.6. Any element can be destroyed and rebuilt at any time without changing what the player sees.

Element	**Position**	**Requirement**
NP counter	top left	Current total with animated gain and spend, plus a cap marker at the 10 NP ceiling from 2.1.1
Player status	top left	Commander portrait shown alongside Core durability (0 to 3)
Enemy status	top right	Mirror of the above for the opponent
Deck and discard	bottom left	Deck count, discard count

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End Turn	bottom right	Enabled only when no queued action remains
Combat log	right edge, collapsible	Last five resolved events in plain language
Targeting overlay	contextual	Highlights valid targets only, matching 2.1.6
Card detail	on hover	Enlarged view with full ability text, dismissible

The HUD honours these constraints:

- No element that reflects a game state change animates longer than the half-second ceiling set in 7.2.
- Damage, healing, and destruction reuse the high-contrast flash defined in 7.2 rather than adding a second visual channel, so the battlefield remains the primary read.
- Cards under Summoning Delay stay dimmed in hand exactly as they are on the battlefield.
- Every displayed value is read from GameState with no local caching, so a rebuilt HUD is always correct.

9. SOUND DESIGN

Audio carries tension, impact, and feedback without competing with the battlefield for attention. The priority order is card feedback first, state changes second, ambience third, and music last. A player who mutes music should still be able to play the entire game correctly.

All audio is sourced rather than commissioned. Assets come from open-source libraries and AI generation, with no original composer. This keeps the audio budget at zero and the MVP achievable on current schedule, but it places a real constraint on the team: every asset must carry a recorded licence and a verified commercial-use position before it enters the build.

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9.1. MUSIC DIRECTION

Music is atmospheric and non-dominant. It supports the fiction of a contested network and never resolves into melody that a player might hum or a turn cycle might fight.

Aspect	**Direction**
Style	Ambient and textural electronic, with sparse melodic elements. Low sustained beds rather than rhythmic drivers, so it does not compete with combat reads
Structure	Layered stems rather than single tracks. Tension builds by adding layers as Core durability falls, not by switching tracks
Intensity mapping	Four intensity levels tied to Core durability (3, 2, 1, 0) plus a game-over variant. The same stem set is reused across all five, so the build stays small
Loop behaviour	All beds loop seamlessly for an indefinite match. No track has a defined end
Relationship to combat	Music ducks approximately 6 dB during combat resolution and returns on resolution, so attack and block sounds stay legible
Restraint	No music plays over the Main Menu or Settings

9.2. MUSIC REQUIREMENT

Track	**Context**	**Length**	**Loop**	**Priority**	**Source**

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					approach
Ambient base bed	All play, neutral state	60 s	Seamless	Must	Open source, modified
Tension layer	Core durability 2 to 1	30 s	Seamless	Must	Open source, modified
Crisis layer	Core durability 0, victory imminent	30 s	Seamless	Should	Open source, modified
Combat layer	Added during attack phase only	20 s	Seamless	Should	Open source, modified
Victory sting	Match won	3 s	No	Must	AI generated
Defeat sting	Match lost or Core destroyed	4 s	No	Must	AI generated
Menu theme	Main Menu only	45 s	Seamless	Could	Open source, modified

All AI-generated stings are re-edited in the DAW and committed as rendered audio, so no runtime generation is used and results never vary between players.

9.3. SFX DIRECTION

Sound effects are short, dry, and functional. Every effect answers one of three questions for the player: what happened, what can I do, or did the game accept my input.

Aspect	**Direction**

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Timbre palette	One shared sample family across factions, so effects read as a single system. Faction identity comes from pitch and layer, not from a distinct sound set per faction
Length	Card and UI effects stay under 400 ms. Nothing that a player must react to exceeds the half-second ceiling set in 7.2
Pitch mapping	Higher pitch for positive outcomes (healing, buff applied), lower for negative (damage, destruction), consistently across every card
Priority	Effects layer rather than cut. Combat resolution events may stack to three voices before the oldest is culled
Consistency	One effect per outcome type, reused everywhere. Damage always sounds the same regardless of which card caused it
Silence as feedback	Invalid input produces no sound rather than an error buzz, matching the input model in 1.5

9.4. SFX REQUIREMENT

Context	**Effect**	**Priority**	**Source approach**
Card play	Card deploys to Active or Bench position	Must	Open source, processed
Card movement	Card moves between zones,	Should	Open source, processed

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	hand to board		
Resource	NP gains at turn start, node control bonus ticks	Must	Open source, processed
Combat	Attack declared, damage dealt, blocker assigned	Must	Open source, processed
Combat	Shield absorbs damage, piercing ignores it	Must	Open source, processed
Status	Buff and debuff applied, duration decremented	Should	Open source, processed
Destruction	Card destroyed, Core durability reduced	Must	Open source, processed
Control state	Node captured, contested, lost	Must	Open source, processed
UI navigation	Menu focus move, confirm, cancel, back	Must	Open source, used raw
UI feedback	Error on invalid action, toggle on and off	Must	Open source, used raw
Match flow	Victory, defeat, rematch offered	Must	AI generated
Ambient	Battlefield room tone, network hum at low intensity	Could	AI generated

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9.5. ASSET PROVENANCE AND LICENSING

Every audio file is committed with a provenance record, and the build is not shippable while any record is missing. This is the main ongoing cost of a no-composer approach and is the reason the provenance table is a hard gate rather than good practice.

Field	**Purpose**
Asset path	Identifies the file in the repository
Source	Library name and URL, or the generation tool and prompt
Licence	Exact licence string as stated by the source, not a summary
Commercial use	Confirmed permitted for a distributed game, with the evidence
Attribution required	Whether attribution must appear in-game or in credits
Modifications	What was changed: pitch, trim, layering, processing
Verified by	Name and date of the person who checked the licence

Two constraints follow from the sourcing decision and need explicit agreement before production work begins.

Open-source licences are not uniform. CC0 permits bundling with no obligation. CC-BY requires visible attribution. CC-BY-SA can impose share-alike terms on derived works, which is the reason modified beds are kept as separate files rather than recombined. Any asset whose licence cannot be resolved to a specific, recorded string is rejected rather than used provisionally.

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AI-generated audio carries a separate risk. Licences for generated output are not settled across all jurisdictions, and platform distribution terms can change independently of copyright law. For that reason AI generation is limited to short stings and ambience where original composition quality matters least, and every generated asset is treated as requiring commercial-use clearance for the target distribution channels before release. If that clearance cannot be obtained, the asset is replaced with an open-source equivalent.

9.6. MIXING AND DELIVERY

The volume controls defined in 8.2 map to three buses: Master, Music, and SFX. SFX is the default bus at full level. Music defaults to 70 percent, and every setting persists across sessions through the save system described in 4.8.

On Web, audio cannot begin until the player interacts with the page, so the first click on the Main Menu initialises the audio context and no sound plays before it. All audio is delivered as compressed streaming assets rather than preloaded files, keeping initial download small for players on slow connections.

10. SCOPE AND MILESTONE

10.1. IN SCOPE

The following features ship in MVP, the first public release. Any feature not listed is deferred beyond MVP; multiplayer and real-money payments require a separate scope decision and production approval.

Core play

Full rules engine: NP economy, six-phase turn structure, Active and Bench slots, summoning delay, node control

Combat resolution with battlefield-local blocking, direct Core attacks, Core Active destruction, and explicit durability-loss events

Win conditions: Core durability reaches zero, deck-out, and concede; Node domination is also implemented per 2.1.8

Card library of 100 cards across four rarities at a fixed 50/30/15/5 distribution, defined as data

Deck construction with faction identity and NP curve validation

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Keyword set as defined in 2.1.11

Presentation

Battlefield view with four nodes per side, Active and Bench positions

Animation, sound hooks, damage popups, and UI highlight feedback

View layer reads snapshots and submits commands only

Opponents and onboarding

- AI Commander using positional heuristics through the shared CommandValidator

- Tutorial match with a restricted rules configuration

Services

Unity Authentication: mandatory online sign-in through Google or another supported identity provider, or as an anonymous guest account

Guest account upgrade path that preserves collection data in place

Cloud Save as the authoritative store for authenticated progression; no offline progression or cache fallback in MVP

Online server-side opening of 5-card packs using the per-slot odds in 2.1.9; earned currency only, no paid currency or real-money purchases

Shipping and quality

PC (Windows) and WebGL builds distributed through itch.io

Free-to-play single-player core; optional 5-card packs use earned in-game currency only. No real-money purchases or paid currency in MVP.

NUnit EditMode and PlayMode suite running against the Core assembly

CardDatabase validation at boot that fails loudly on malformed definitions

10.2. OUT OF SCOPE

Listed explicitly so that scope creep is a decision rather than an accident.

Mobile and tablet ports, including touch UI and app store distribution

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Direct online PvP (including same-platform matches), matchmaking, and cross-platform play; all deferred beyond MVP pending a separate approved milestone.

Ranked seasons, matchmaking rating, and any form of automated matchmaking

Player trading, card auctions, or any secondary market

Open sandbox mode where players author their own cards

Modding support or scripting hooks exposed to players

More than three factions in the MVP; additional factions are planned for post-MVP

Player chat and voice communication

Cinematics, voice acting, or branching narrative beyond card flavour text

Physical print run or merchandise

Anti-cheat measures beyond deterministic server-side validation and replayable logs

10.3. MILESTONES

10.3.1. PRE-PRODUCTION

Card schema frozen and the JSON export pipeline working end to end

asmdef boundaries established, with Core compiling as a plain .NET class library

CommandValidator and EffectInterpreter covered by unit tests

Twenty-card playtest set drafted and playable in an internal build

Save schema v1 written, with the migration path stubbed

10.3.2. PRODUCTION

Rules engine complete: phases, economy, combat, and victory conditions

Card library authored to target count and validated by CardDatabase at boot

Battlefield UI with animation and audio integration

AI Commander heuristic scorer tuned against recorded playtest sessions

Tutorial flow implemented from first launch to first completed match

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Cloud Save and Authentication integrated and verified on both platforms

10.3.3. TESTING

Core assembly coverage on validator, interpreter, and combat paths
Validator fuzz-tested against malformed and out-of-order command sequences
External playtest builds with a maintained balance spreadsheet
Save migration verified across every schema version in use
Pack opening odds verified server-side, with results logged before response
Regression pass across both the PC and WebGL targets

10.3.4. RELEASE

Legal, ratings, and platform compliance check for free-to-play distribution itch.io page, store listing, and first-look assets prepared
Crash-free session target met on low-to-mid-range hardware
Release candidate verified on PC and WebGL
Post-launch hotfix process documented and rehearsed

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11. DEVELOPMENT REQUIREMENTS

Technical and organizational groundwork. Split into:

11.1. TECHNICAL SPECIFICATIONS

Specification	**Details**
Game Engine	Unity
Programming Language	C#
Target Platforms	Windows PC and WebGL
Distribution Platform	itch.io
Display Resolution	1920 × 1080 for PC, with responsive scaling for WebGL
Performance Target	60 FPS on recommended hardware, with a minimum target of 30 FPS on low-end hardware
Rendering	2D rendering using Unity's 2D systems
Networking	MVP: local single-player matches on Windows PC and WebGL. Direct online multiplayer and its client-server/host architecture are deferred beyond MVP.
Data Storage	JSON and Unity-compatible serialized data for card, deck, and game configuration data
Image Formats	PNG for sprites, card artwork, UI elements, and other 2D assets
Audio Formats	WAV for source audio and compressed formats such as OGG for in-game audio

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Build Format	Windows executable build and WebGL build for browser deployment

11.2. THIRD PARTY TOOLS

Tool	**Purpose**
GitHub	Source code hosting and version control
Git	Local version control and development history management
Figma	UI/UX design and interface prototyping
Google Docs	Game documentation and project planning
Generative AI Tools	Generation and assistance in creating 2D artwork, card illustrations, visual assets, and other supporting game assets
Visual Studio Code	Code editing and development

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12. MONETIZATION

Overlink MVP is free-to-play and distributed through itch.io for PC and Web. Players can access the single-player core without an upfront purchase; the first public release includes no real-money purchases or paid currency.

The MVP card-collection loop offers randomized 5-card packs purchased only with in-game currency earned through play. Pack rarity odds and selection are defined in 2.1.9. Real-money purchases, paid currency, and payment-provider integration are deferred beyond MVP and require a separate scope, platform, and compliance review.

MVP progression must remain playable without purchases: all pack currency is earned through play, and no paid advantage or paid acquisition ships. Any post-MVP proposal for real-money purchases requires new approval and a separate review of odds disclosure, platform/payment rules, and accessibility.

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13. RISK ASSESSMENT

Category	Risk	Impact	Mitigation Plan
Schedule	Development takes longer than the available semester timeline.	High	Prioritize the core gameplay loop and establish a minimum viable product (MVP). Defer non-essential features until the core game is stable.
Scope	Too many cards, mechanics, factions, or features increase development complexity.	High	Define a fixed MVP scope and limit the initial card pool, mechanics, and game modes.
Technical	Post-MVP risk: online multiplayer is more difficult to implement than expected; it is explicitly excluded from the first public release.	High	Do not let multiplayer delay MVP. Only if a later multiplayer milestone is approved, prototype transport and the authoritative command path before producing multiplayer-specific content.
Technical	WebGL performance or browser	Medium	Test WebGL builds regularly on different

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	compatibility issues affect the Web version.		browsers and optimize assets, scripts, and rendering where necessary.
Technical	AI opponent behaves poorly or makes ineffective decisions.	High	Start with a simple rule-based or utility-based AI and gradually improve its decision-making through testing.
Content	Insufficient cards or visual assets are completed before the deadline.	Medium	Prioritize cards required for the MVP and use generative AI to accelerate the creation of supporting visual assets.
Game Design	Card balance becomes difficult to maintain as the card pool grows.	High	Use clearly defined card roles and statistics, then conduct repeated playtesting to identify overpowered or underpowered cards.
Multiplayer	Synchronization or connection problems cause inconsistent game states between	High	Post-MVP only: keep online interactions simple and validate every

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	players.		game-state change on the server. MVP release acceptance does not depend on multiplayer.
Quality	Bugs in card interactions or game rules cause unexpected behavior.	High	Implement gameplay systems in modular components and perform regular playtesting and regression testing after major changes.
Solo Development	Managing programming, design, documentation, testing, and assets alone becomes overwhelming.	High	Maintain a prioritized task list, work in small milestones, and focus on completing essential features before optional improvements.
Wellbeing	Excessive workload or poor time management reduces productivity and affects project completion.	High	Set realistic weekly goals, maintain a development schedule, and reserve time for rest and academic responsibilities.
Technical Debt	Quick implementations	Medium	Use modular architecture,

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	made to meet deadlines make later changes difficult.		maintain basic documentation, and periodically refactor critical systems when time permits.

14. TESTING AND QA PLAN

Testing Approach

Testing runs continuously, with each major gameplay feature tested after implementation. Playtests cover card interactions, combat, AI, progression, account/cloud services, and online pack openings. Online multiplayer is post-MVP and is not an MVP test or release gate. Before release, verify the single-player game and services on PC and WebGL.

Types of Testing

Test Type	Description
Functional Testing	Verify that cards, abilities, combat, resources, turns, Node control, and victory conditions work as intended
Internal Playtesting	Conduct regular solo playtesting to identify gameplay issues, usability problems, and unexpected interactions
AI Testing	Test whether the AI can correctly evaluate game states and perform valid actions without becoming stuck or making illegal moves
Balance Testing	Test card statistics, abilities, resources, and faction interactions to identify excessively strong or weak strategies

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Post-MVP Multiplayer Testing	Only if a separate post-MVP multiplayer milestone is approved: test matchmaking, synchronization, turn handling, disconnections, and server-authoritative state. Not an MVP test or release gate.
Platform Testing	Test Windows and WebGL builds across supported browsers and hardware configurations
Regression Testing	Retest existing systems after major updates to ensure new changes do not break previously working features

Bug Tracking

Bugs will be recorded using GitHub Issues. Each issue will include a description of the problem, reproduction steps, affected system, severity, and current status. Bugs will be prioritized based on their impact on gameplay, with critical gameplay-breaking issues addressed before minor visual or usability issues.

QA Priorities

The following systems must function correctly before release:

1. Core gameplay loop and turn progression.
2. Card deployment, combat, abilities, and Active-Bench interactions.
3. Resource management and victory conditions.
4. Node and Core interactions.
5. AI opponent functionality.
6. For any separately approved post-MVP multiplayer milestone: synchronization and server-authoritative game-state consistency.
7. Card collection and gacha functionality.
8. Saving and loading of relevant player data.
9. Stable PC and WebGL builds.

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10. Critical UI interactions and error handling.

Limitations

Due to the limited development time and solo development structure, exhaustive testing across hardware, browsers, card combinations, and service failures is not possible. MVP testing prioritizes single-player rules, AI, supported PC/WebGL builds, account/cloud-save paths, and online pack transactions. Multiplayer is deferred, not an untested MVP feature. Minor visual inconsistencies and rare edge cases may remain for later updates.

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